

Sardine Stock Assessment Model

International Stock Assessment Workshop

Cape Town

2-6 December 2024

Carryn de Moor

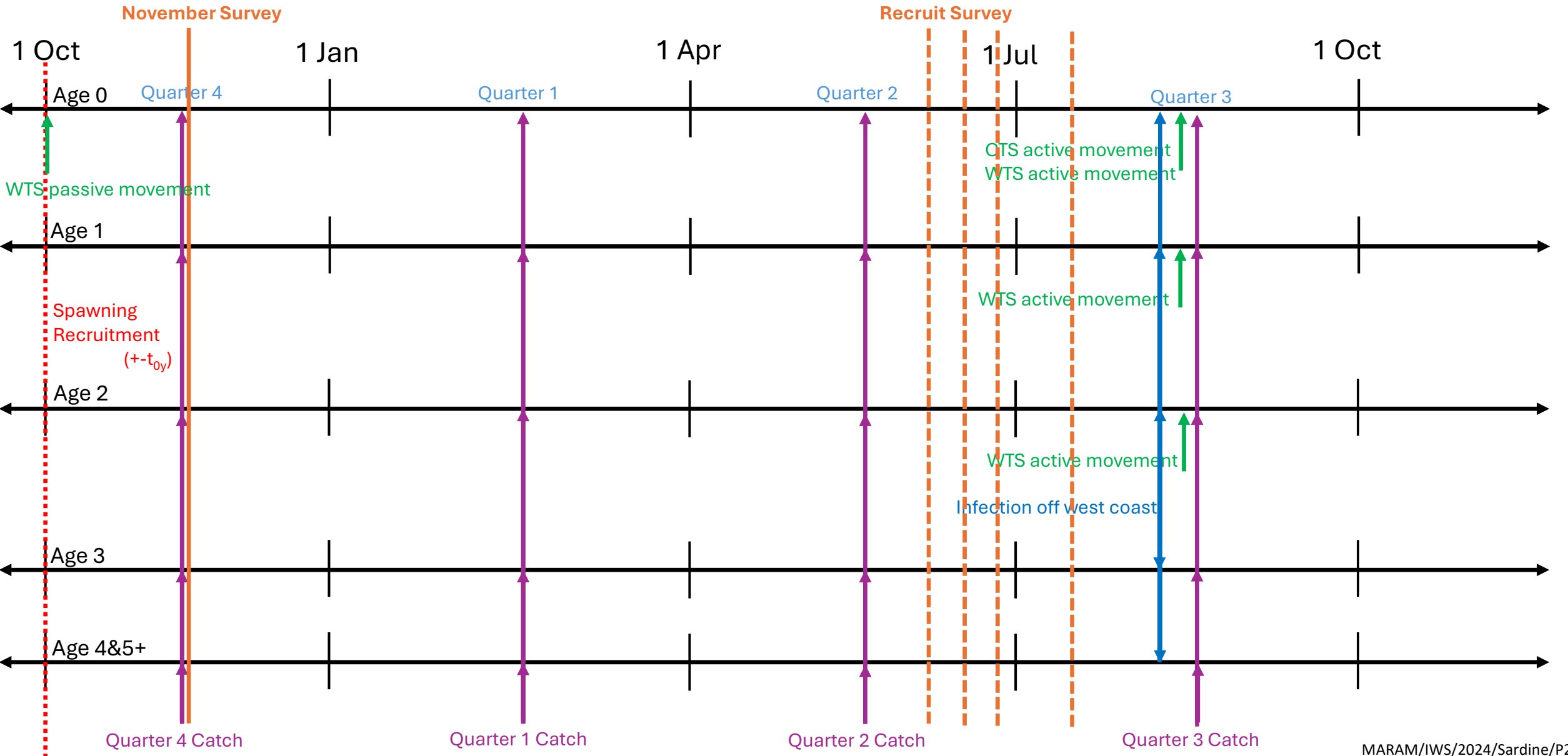


Marine Resource Assessment and Management Group (MARAM)
Department of Mathematics and Applied Mathematics
University of Cape Town

Sardine stock assessment model

- Two-areas (split at Cape Agulhas)
- Two stocks (CTS and WTS)
- Quarterly catch is modelled to be taken in a pulse at the mid-point of each quarter.
- Recruitment of WTS and CTS off the west coast are modelled independently.
- No CTS recruitment off the south coast
- Spawning / recruitment are assumed to occur at the start of the 4th quarter + allowance for earlier/later peak recruitment (1st October).
- It is expected that the timing of recruitment will be estimated to be earlier for the south coast, and with faster growth in the first few months on the south coast than the west coast (Sakamoto *et al.* 2020, van der Lingen *et al.* 2023).

Timeline of key stock assessment processes

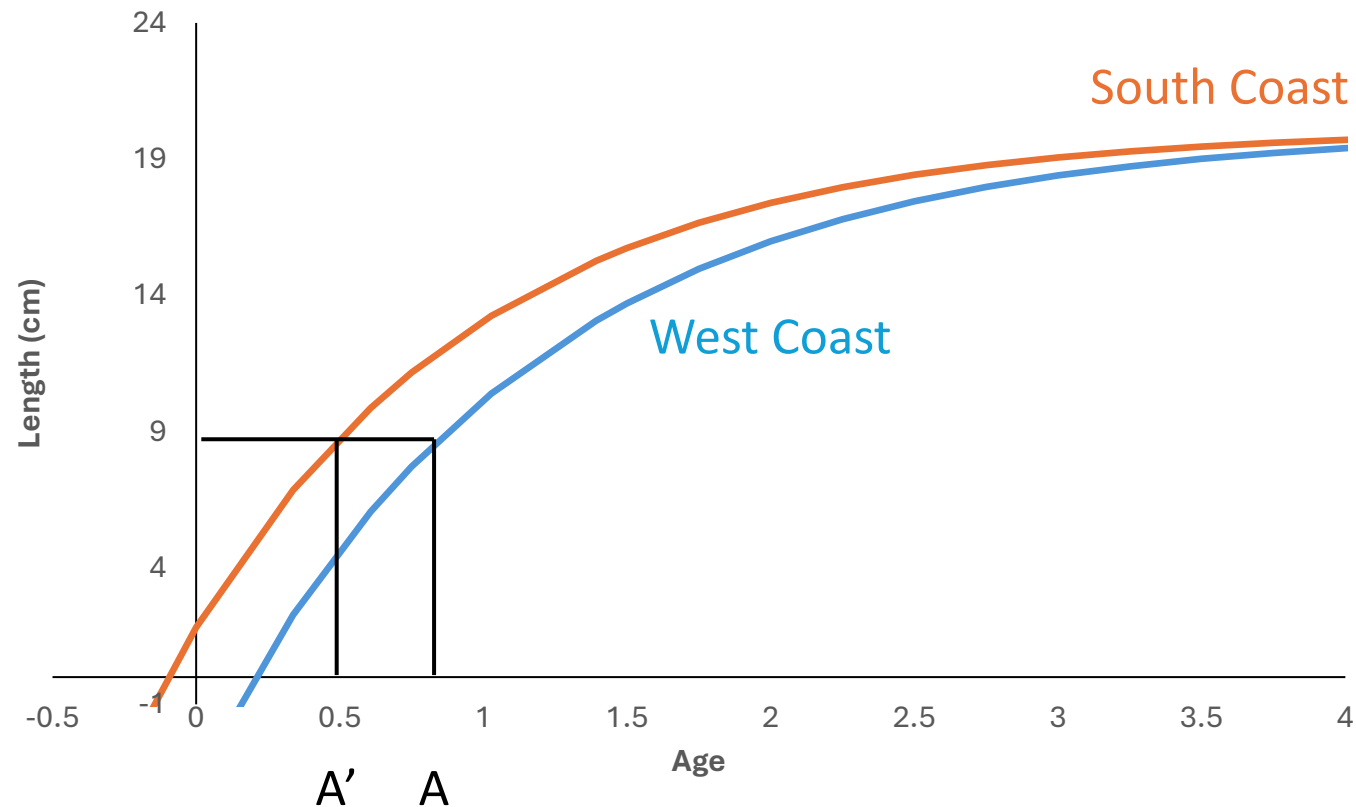


Sardine stock assessment model

- Annually-varying proportion of WTS spawned off the south coast are passively transported to the west coast as recruits.
- Active movement of WTS (age <1 - <3) and CTS (age <1) from the west to the south coast is assumed to occur at some point after the recruit survey and before the November biomass survey (van der Lingen *et al.* 2023). (3rd quarter)
- Parasite infection is assumed to occur in bursts as parasites are released from their subtidal first intermediate host (de Moor *et al.* 2017). This is currently proposed to occur annually at the middle of the 3rd quarter, before active movement. This implies that infected sardine could reach the south coast from 9 months of age.
- Both hydroacoustic surveys are assumed to provide relative indices of abundance.
- At a later stage, some WTS may need to be modelled to remain on the west coast for a longer period if this assumption does not allow the model to fit the increasing parasite prevalence-at-length rates observed off the south coast.

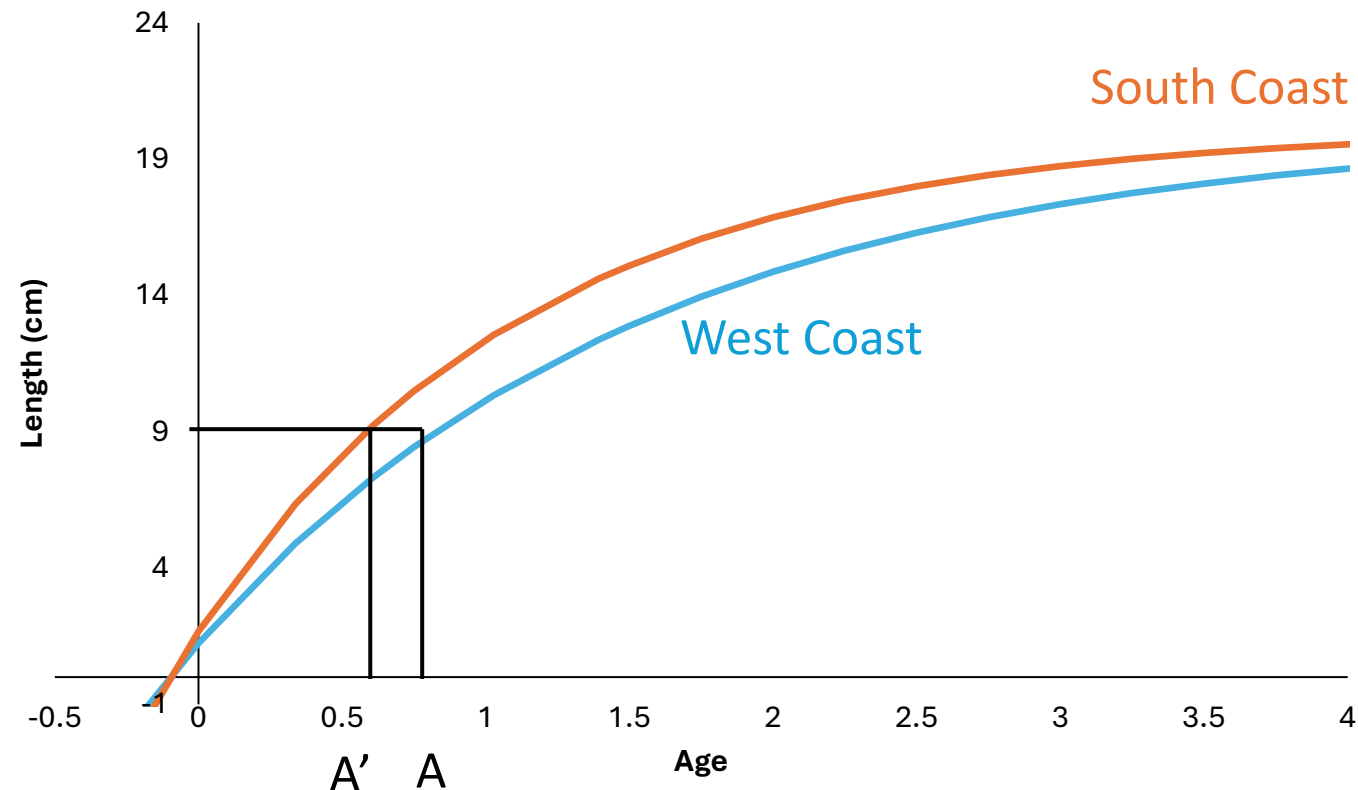
Sardine stock assessment model

- Growth curves of WTS that move from the west coast to the south coast reflect their growth according to the west coast for the first part of their life and then after they move to the south coast, at the same rate of fish off the south coast of the (lower) age - that corresponds to their length at time of movement (see Appendix B).

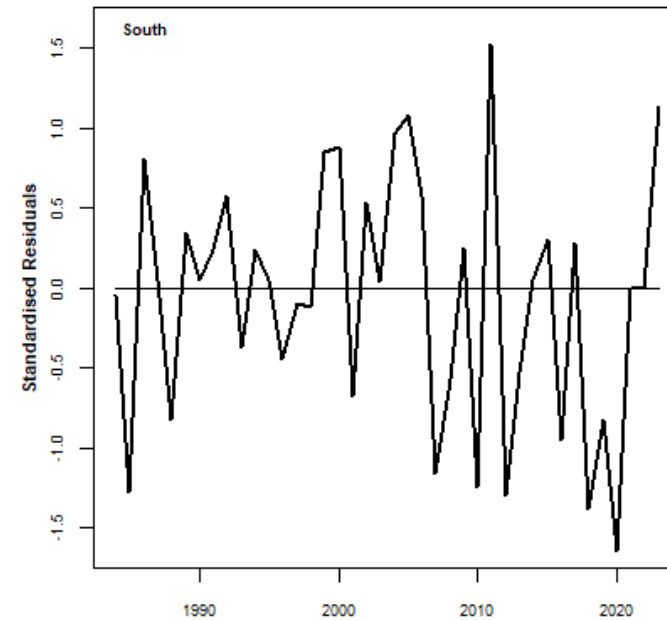
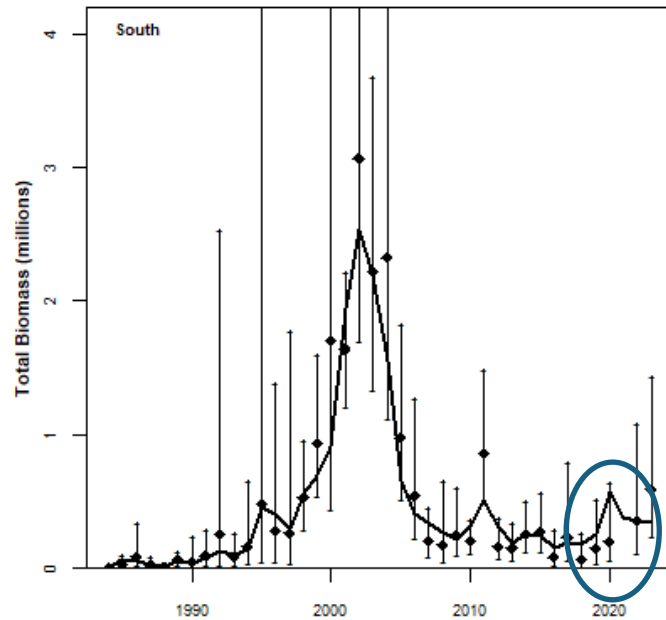
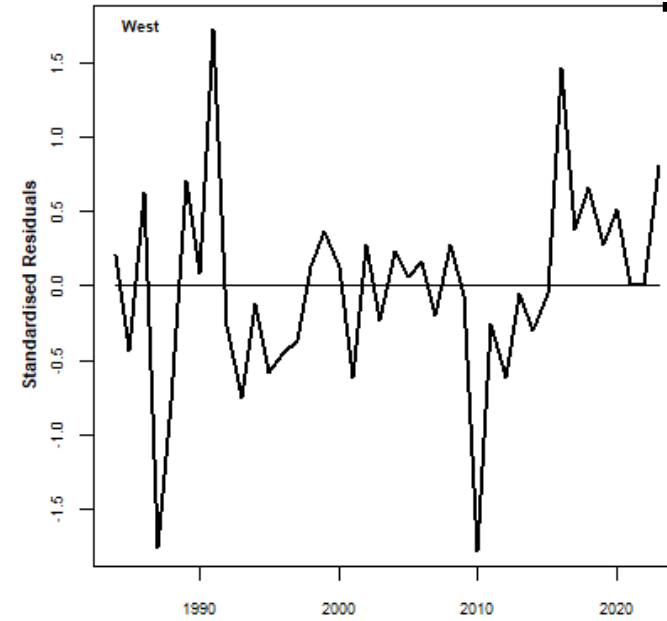
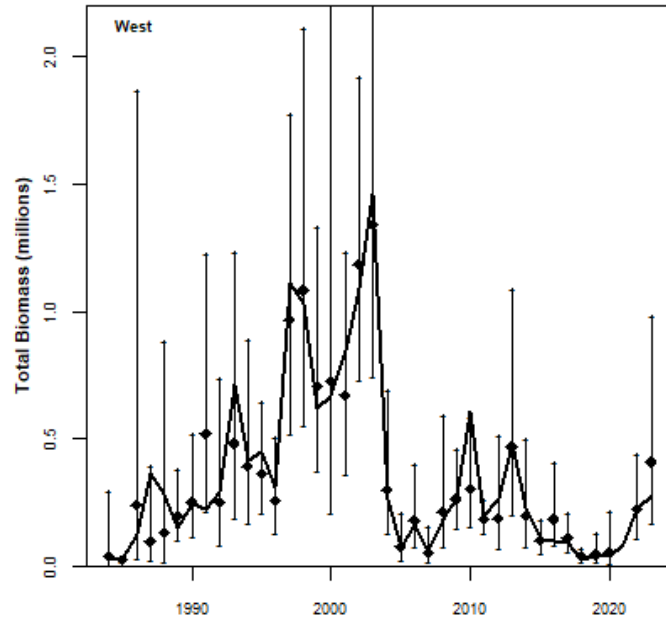


Sardine stock assessment model

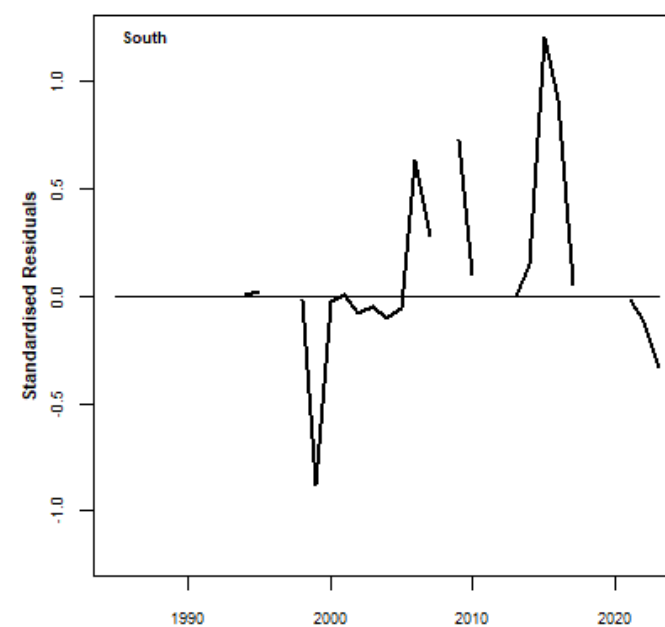
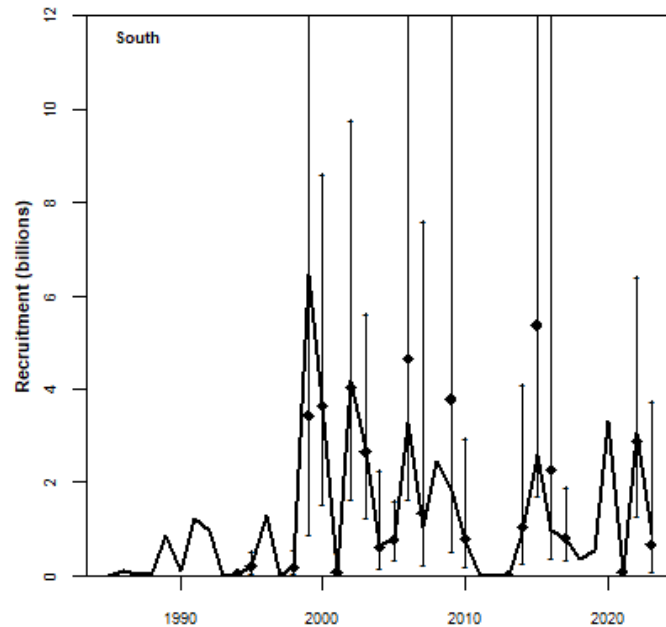
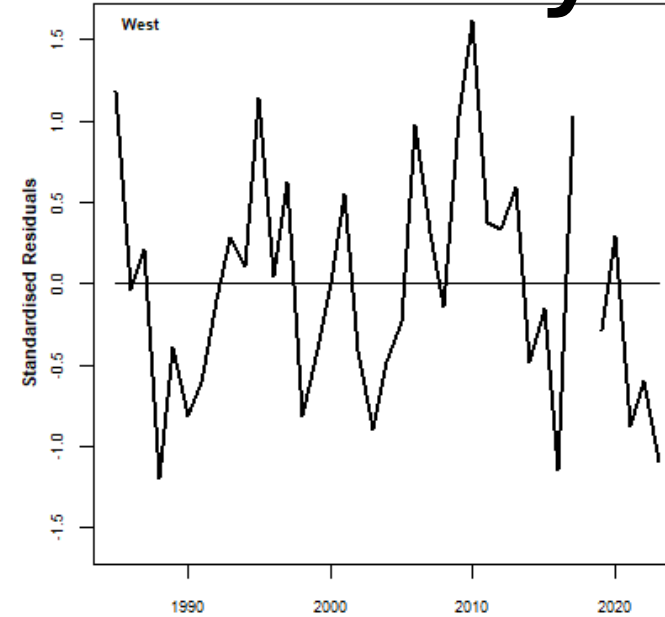
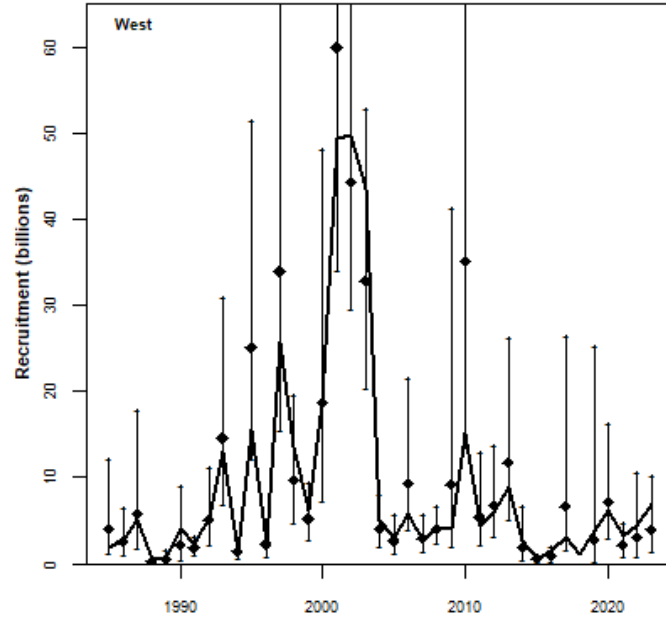
- Growth curves of WTS that move from the west coast to the south coast reflect their growth according to the west coast for the first part of their life and then after they move to the south coast, at the same rate of fish off the south coast of the (lower) age - that corresponds to their length at time of movement (see Appendix B).



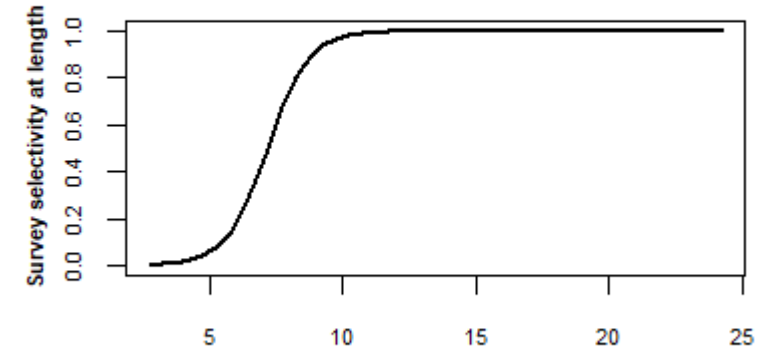
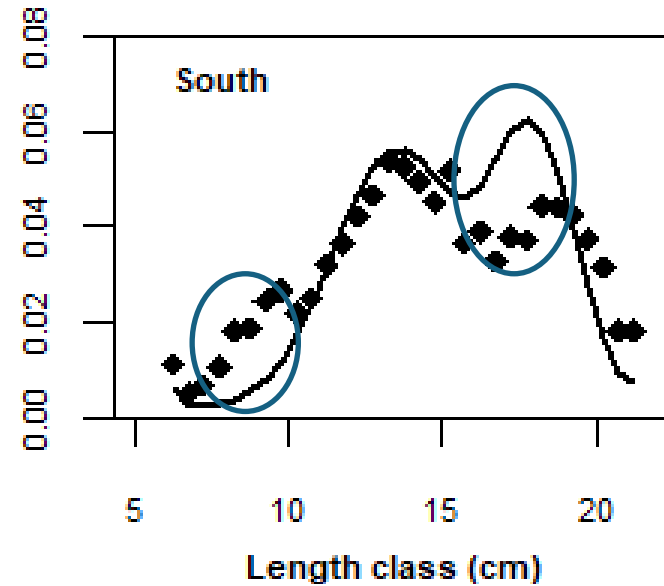
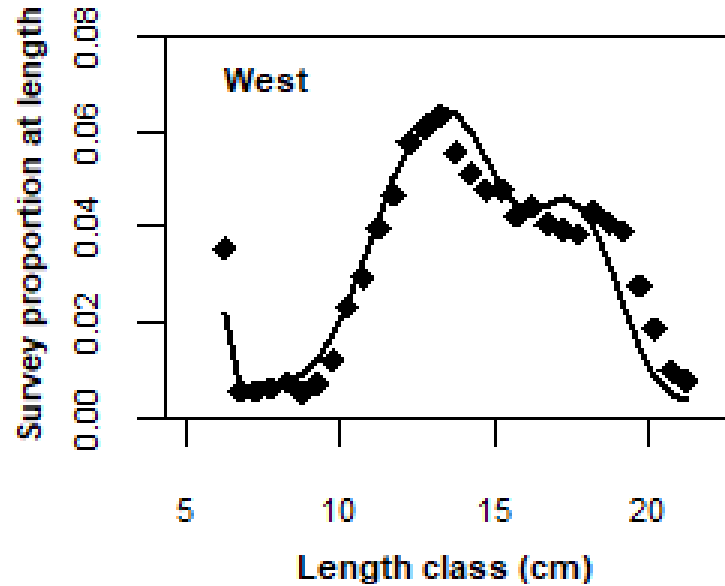
Model fit to November survey data



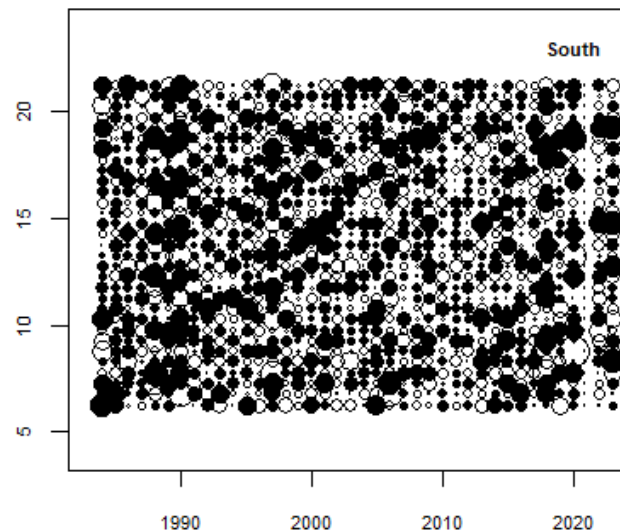
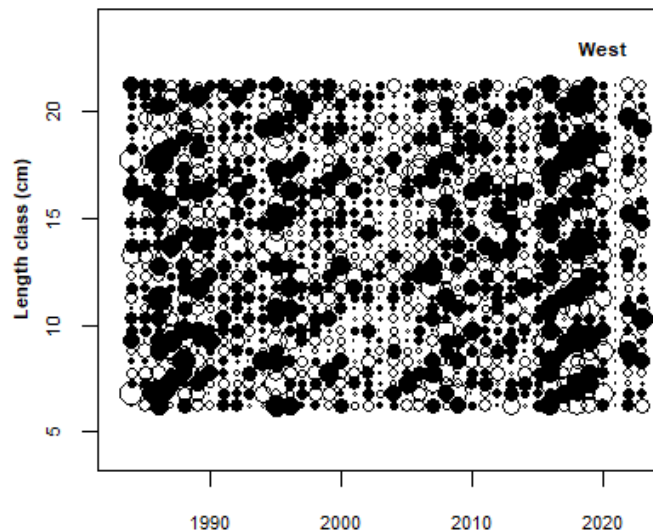
Model fit to recruit survey data



Model fit to November length frequencies



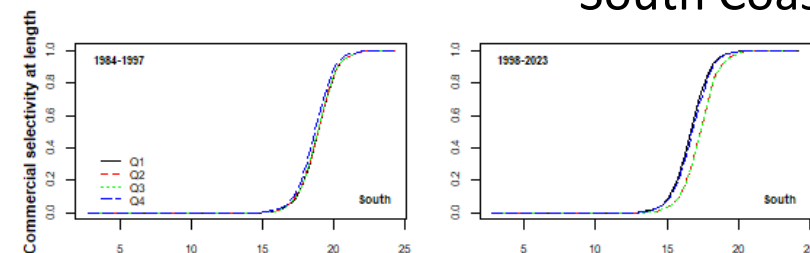
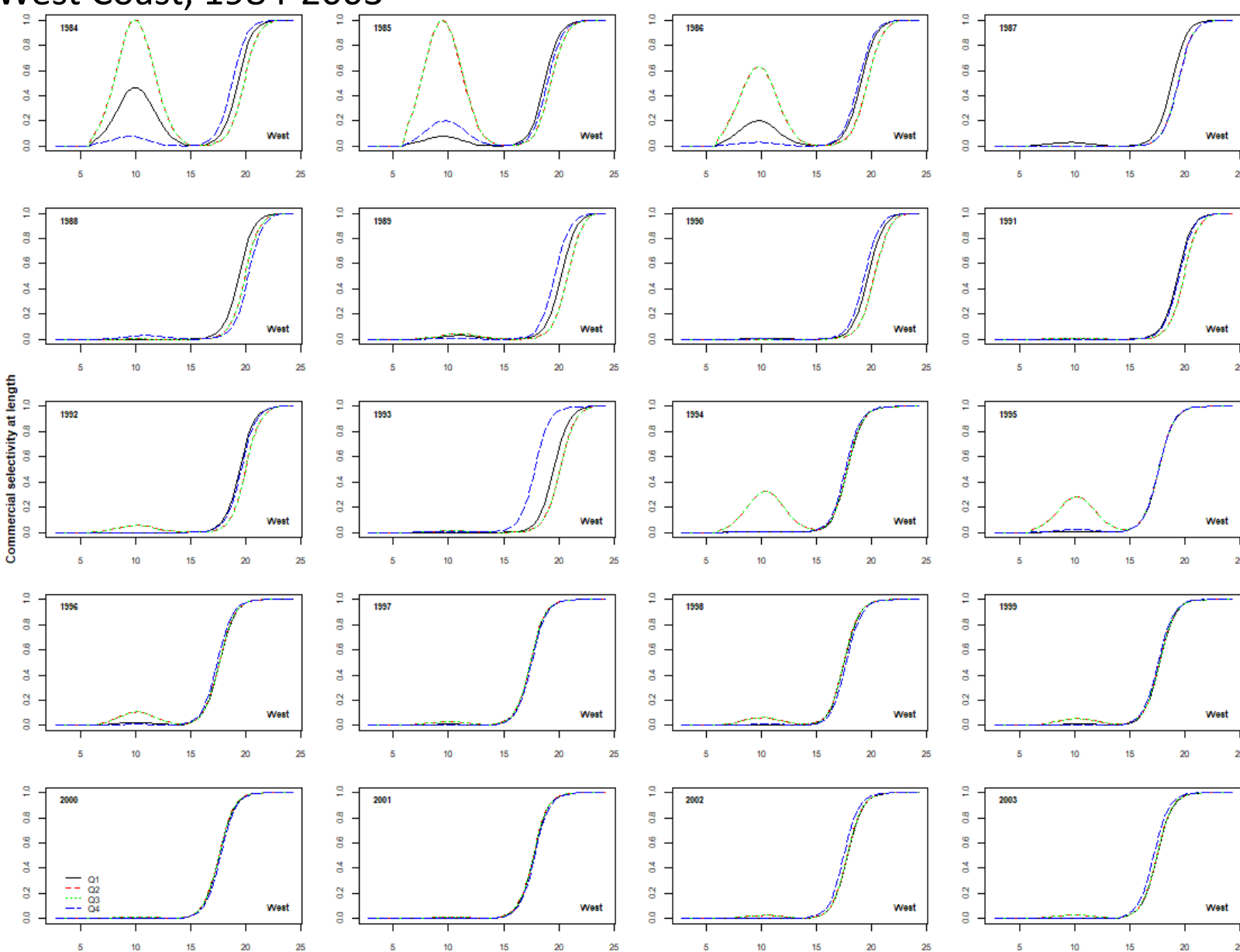
$$\text{Sel}_{50} \sim 1/(1+e^{(S_{50}-7)/0.5})$$
$$\text{delta} \sim U(0.005, 5)$$



Commercial selectivity

West Coast, 1984-2003

South Coast



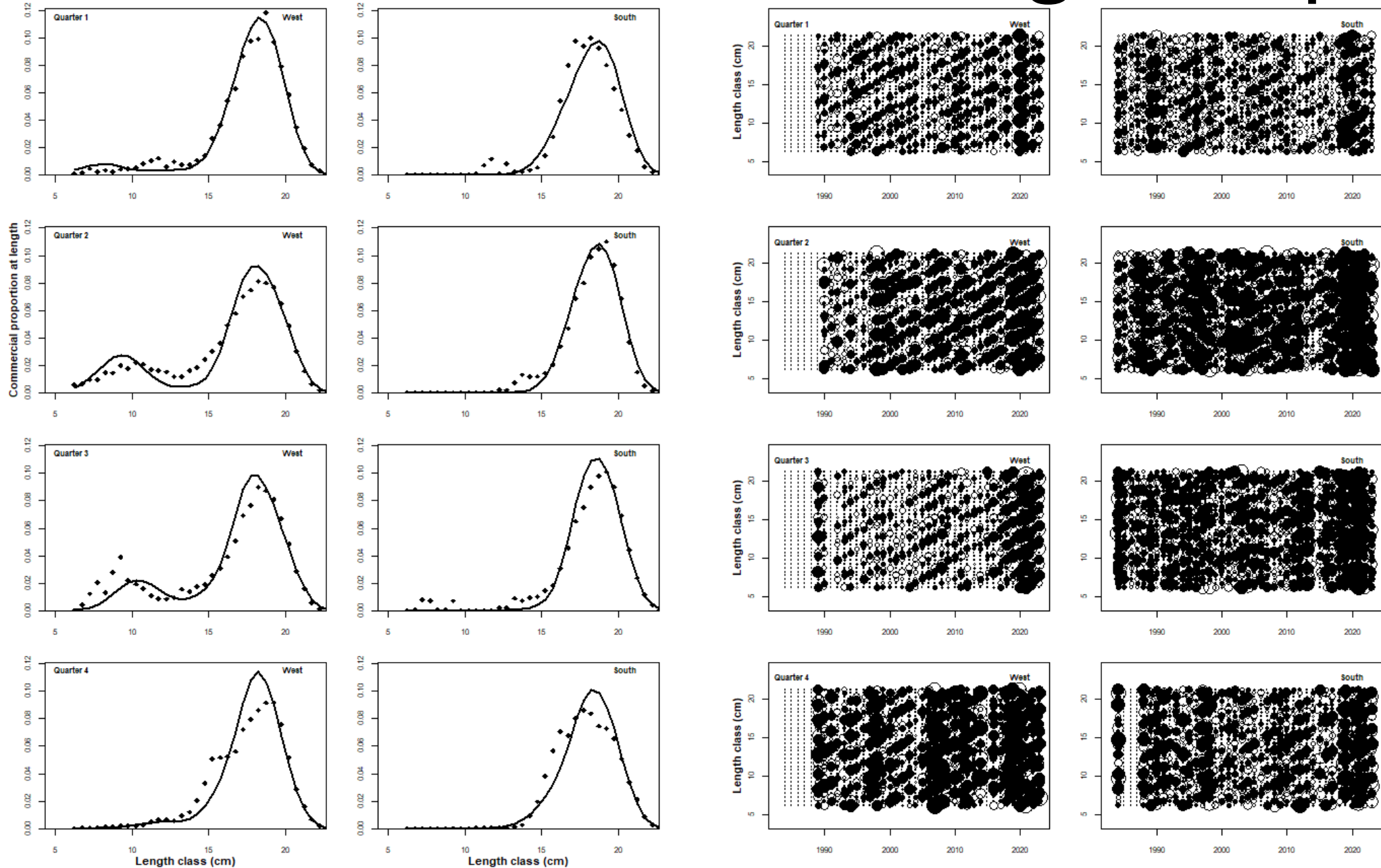
No normal dist for small
bycatch on south coast

$l_{1,y} \sim N(10, 1^2)$
 $l_{2,c,y,q} - l_{1,y} \sim U(0, 15)$
 Estimated for two quarters
 and two year time periods

$\text{Sig}_1^2 \sim U(2, 15)$
 $\text{Sig}_2^2 \sim U(0, 10)$

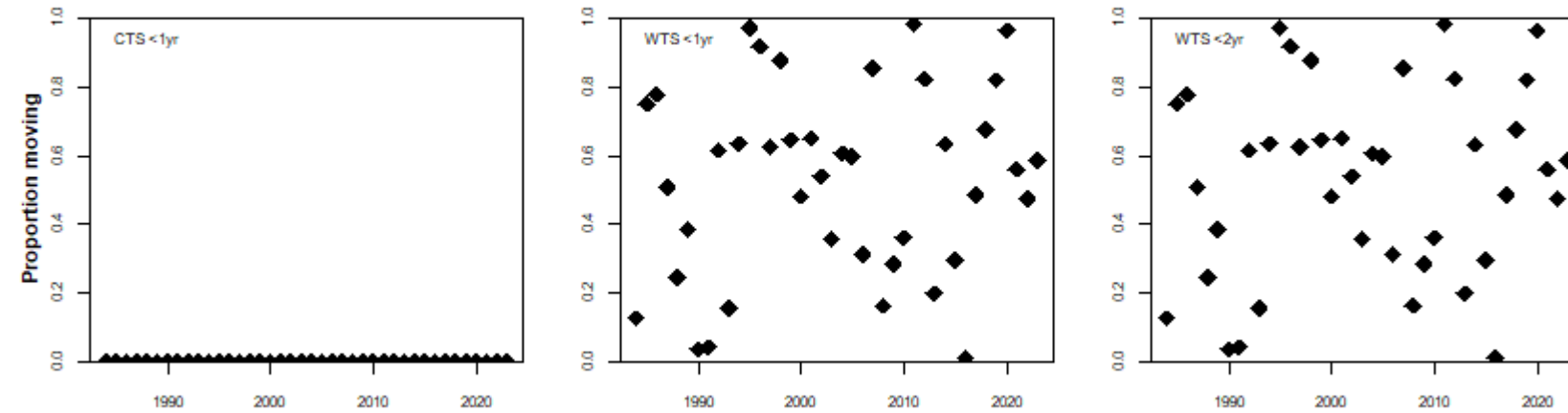
$S(c, y, q=1, a) = S(c, y-1, q=4, a)$
 $S(c, y, q=2, a) = S(c, y, q=3, a)$

Model fit to commercial length frequencies

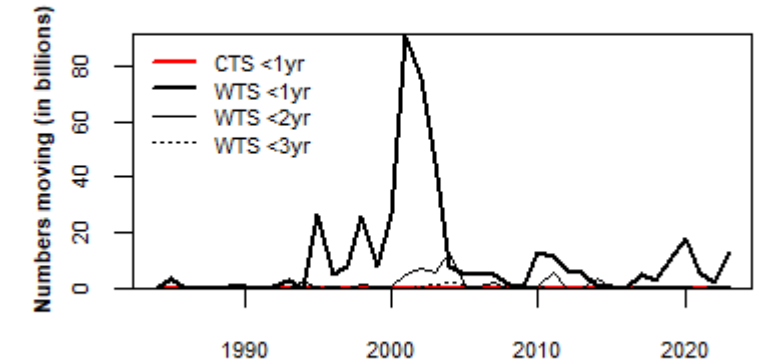


Movement

Active Movement – Proportions

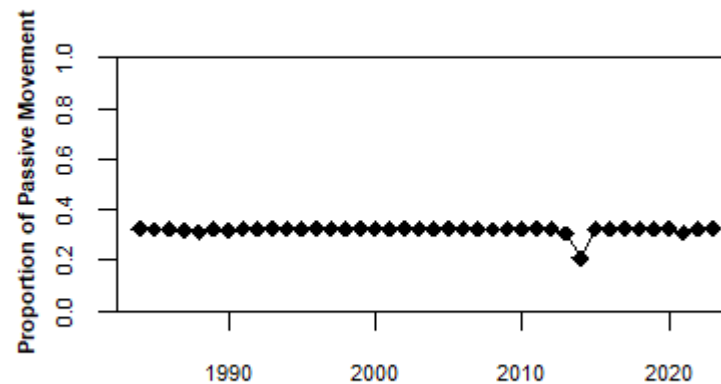


Active Movement - Numbers



$$\begin{aligned} \text{move}_0^{\text{CTS}} &\sim 1/(1+e^{(\text{move0CTS}-0.1)/0.1}) \\ \text{move}_0^{\text{WTS}} &\sim \text{Beta}(1.05, 1.05) \\ \text{move}_1^{\text{WTS}} &= \text{move}_0^{\text{WTS}} * U(0,1) \end{aligned}$$

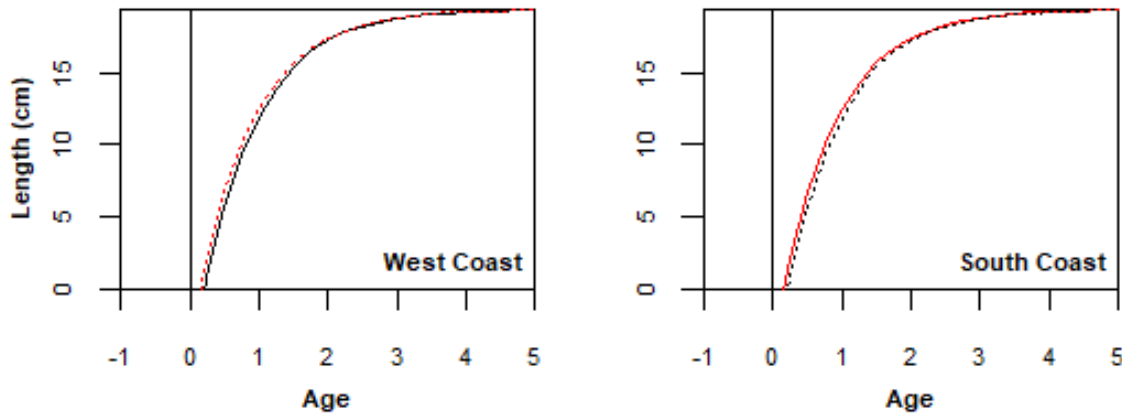
Passive Movement



$$p_y \sim N(0.325, 0.165^2)$$

Growth

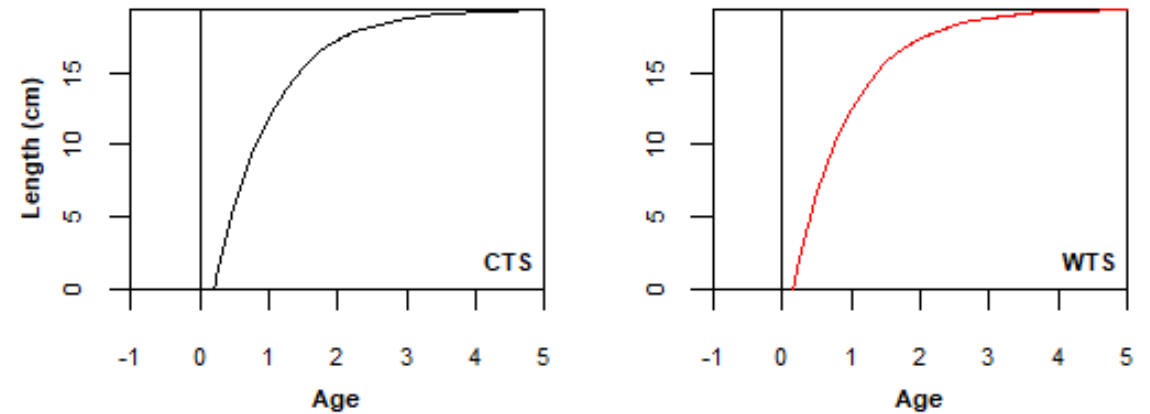
Von Bertalanffy by coast



$t_0(\text{WTS},w)=t_0(\text{WTS},s) \sim U(-\infty,\infty)$
 $t_0(\text{CTS},s)=t_0(\text{CTS},w) \sim U(-\infty,\infty)$
 i.e. timing of spawning is stock-specific

Solid line – preferred habitat
 Dotted line – other coast

Von Bertalanffy by stock



$L_{w,1} \sim U(5,18)$
 $L_{s,1} - L_{w,1} \sim U(0,13)$
 i.e. length at age 1 greater on SC

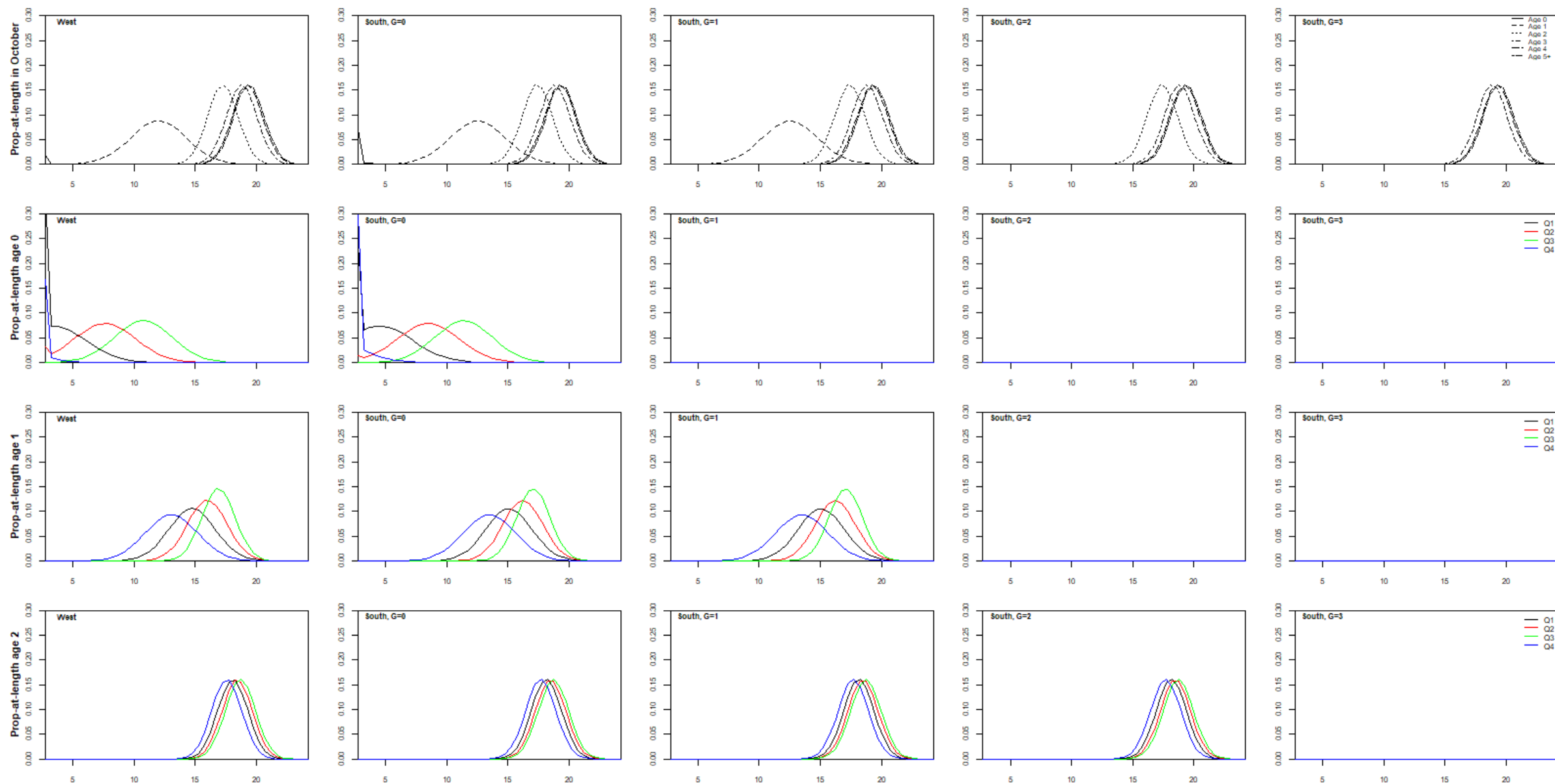
$\kappa_w \sim U(0,2)$
 $\kappa_s - \kappa_w \sim U(0,2)$
 i.e. shape parameter should allow for faster SC growth

Additional Constraint: $L_{w,\infty} \leq L_{s,\infty}$

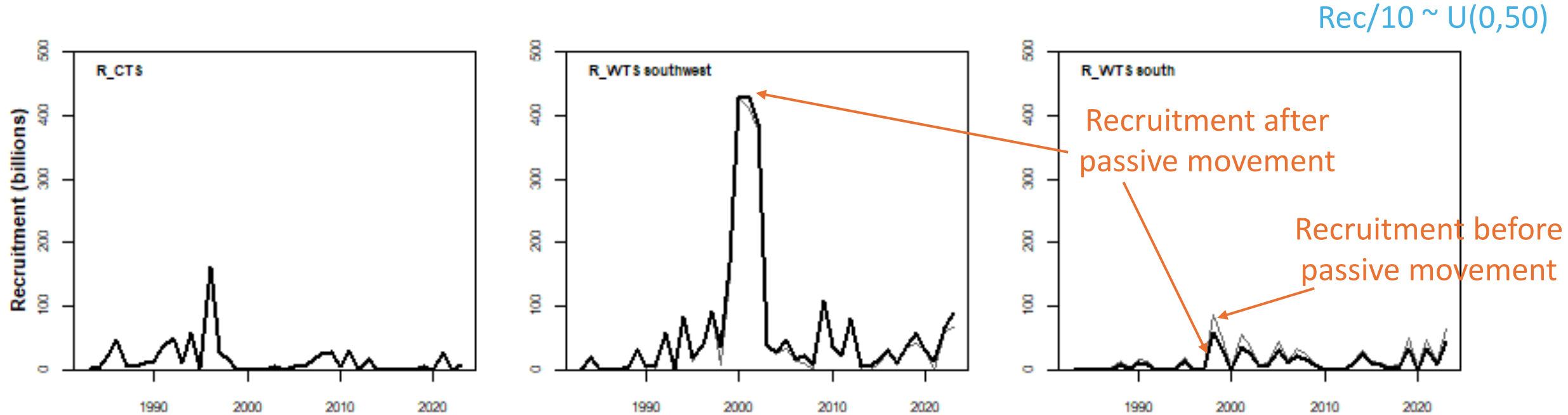
<1 month difference between peak recruitment of SC and WC; peak recruitment of SC ~ late November

Sig/3 ~ Beta(1.05,1.05)
i.e. almost U(0,3)

Length-at-age



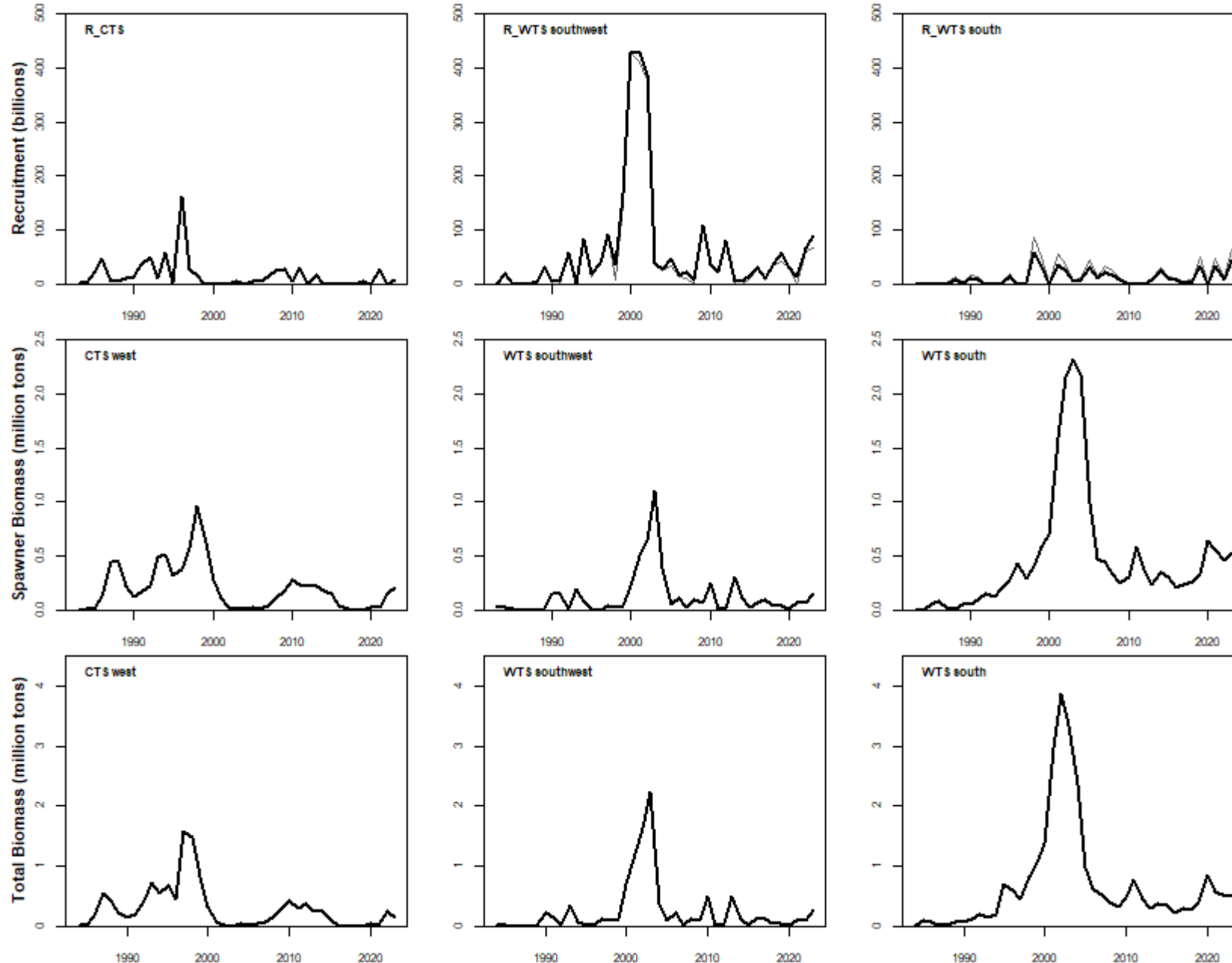
Recruitment



FISHERIES/2024/NOV/SWG-PEL/41 tested alternative priors on passive movement

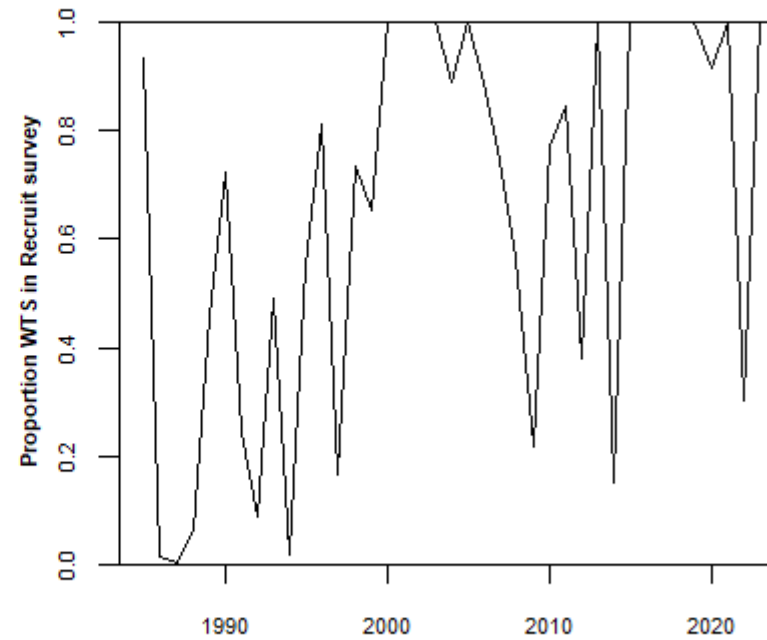
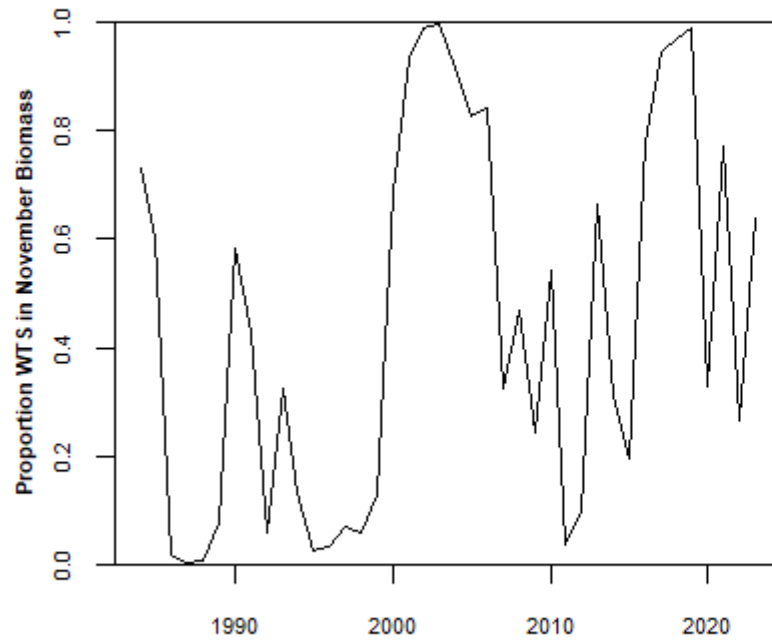
- * Without priors, passive movement does not impact fit to data, but is important for recruitment – SSB comparison by coast

Time Series



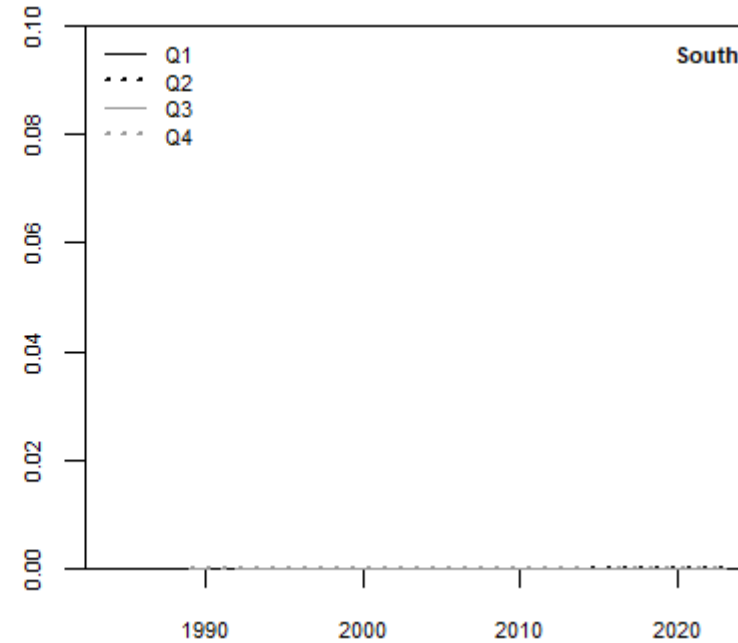
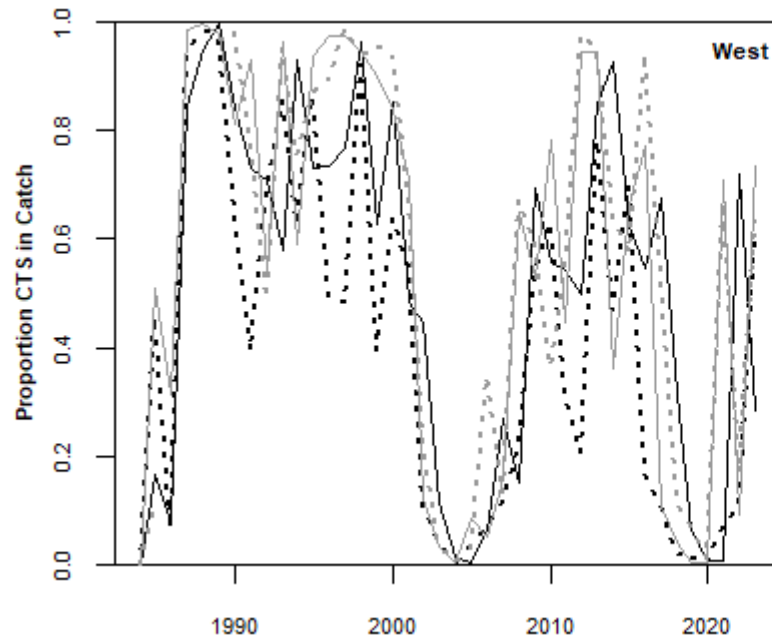
WTS > CTS off west coast in many years, not only those corresponding to the pulse at the turn of the century

Time Series

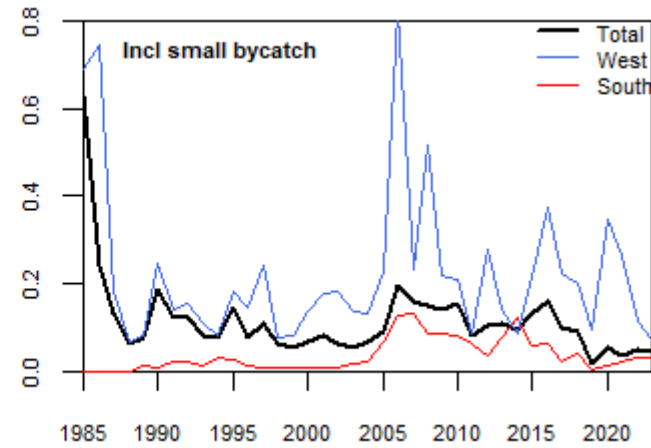
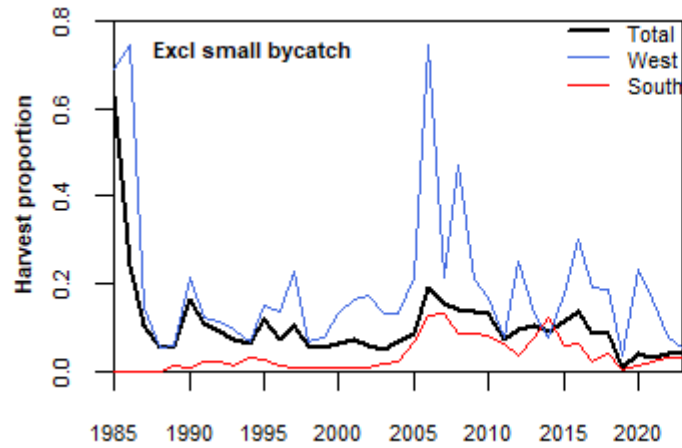


WTS > CTS off west coast in many years, not only those corresponding to the pulse at the turn of the century

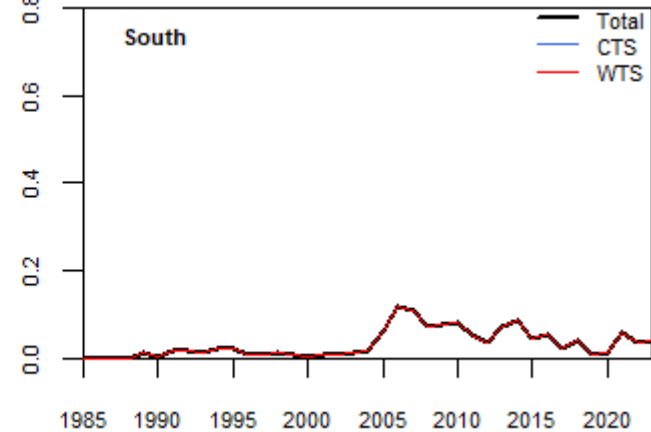
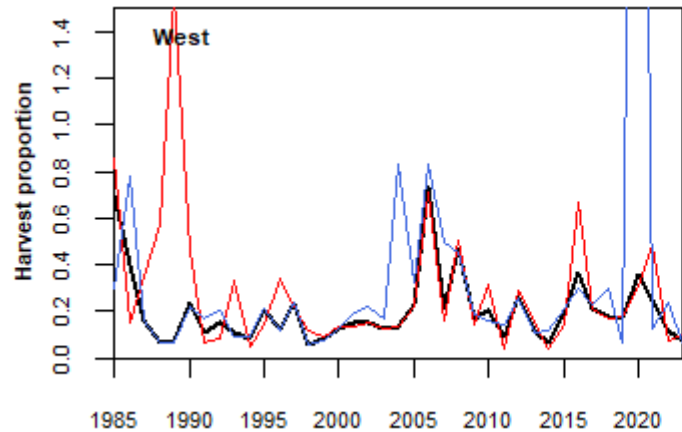
CTS in commercial catches



Historical harvest proportions

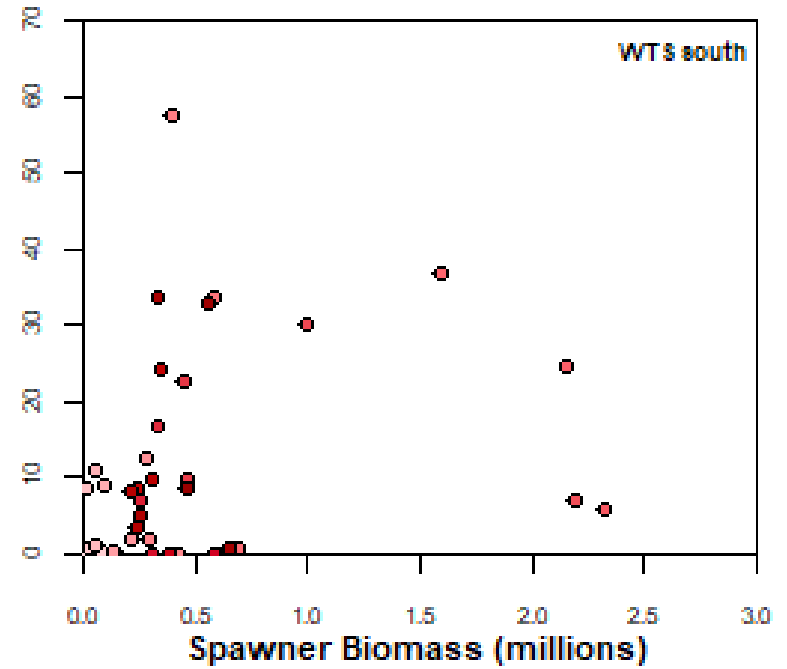
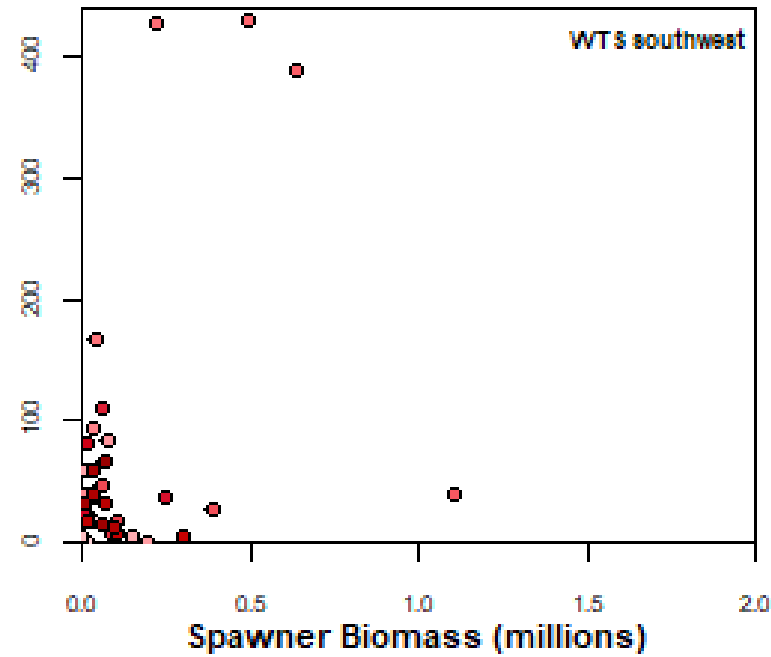
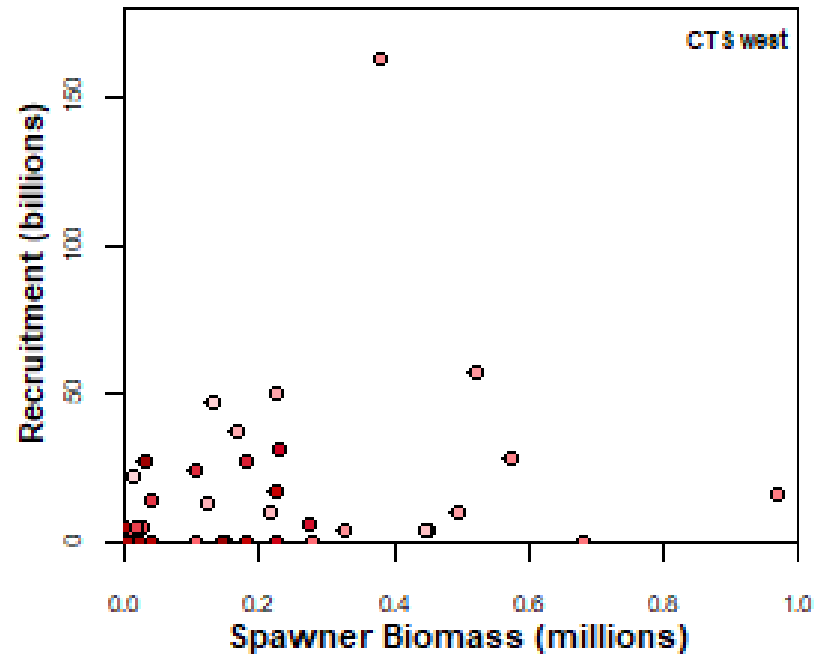


(Jan-Dec catch tonnage) /
November biomass

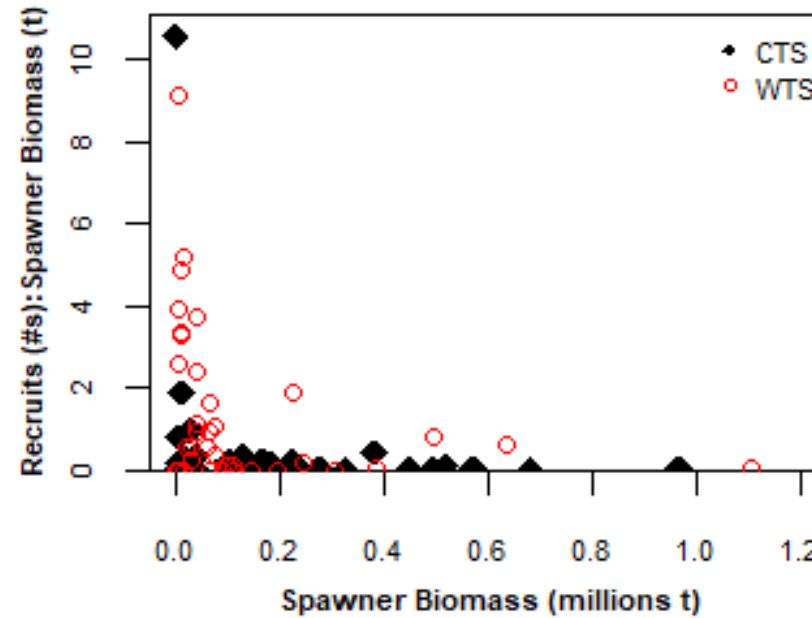


(Catch ($N_c^j * w_l$) summed over 4 quarters)) /
November biomass by stock, coast

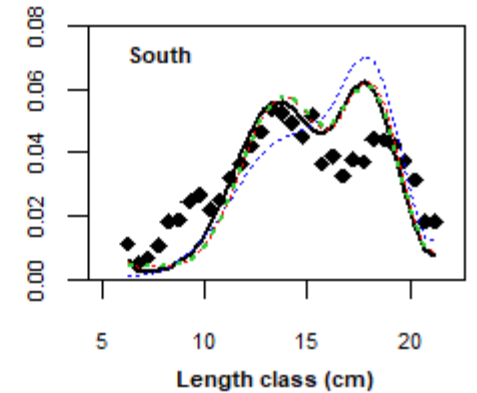
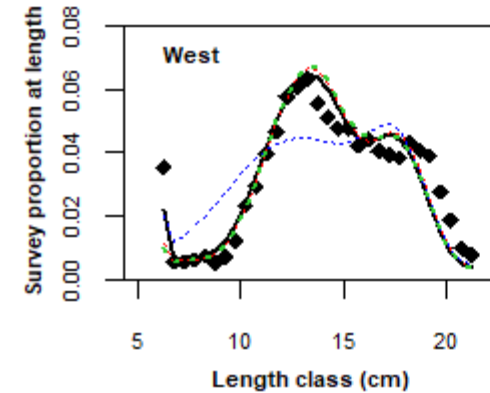
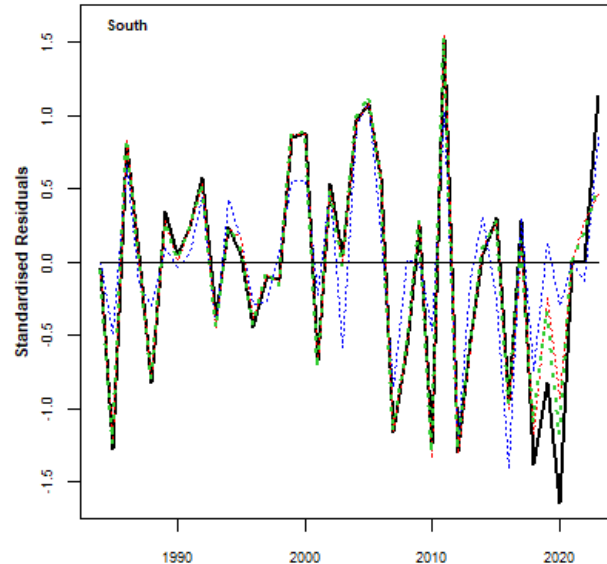
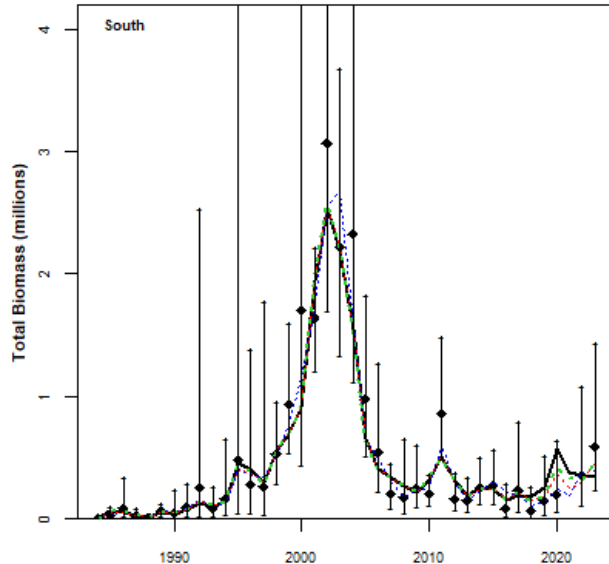
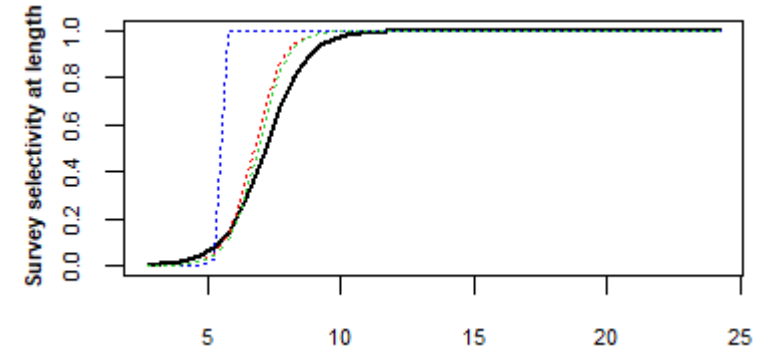
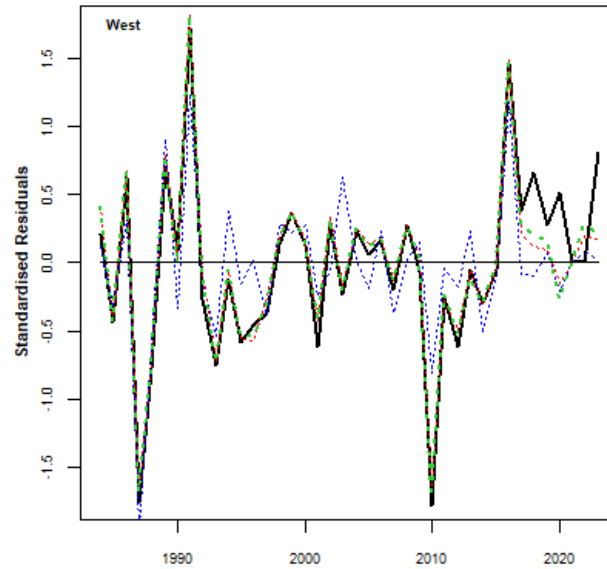
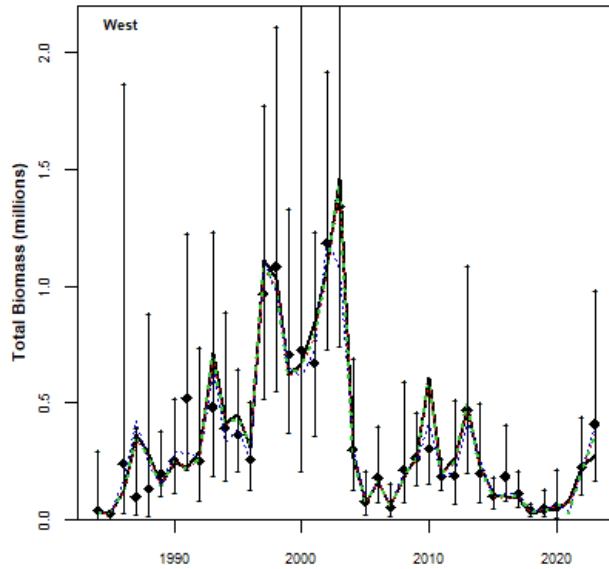
Stock-Recruitment scatter plot



Recruits:Spawners v Spawner biomass



Improving fit to Nov survey – data conflict?

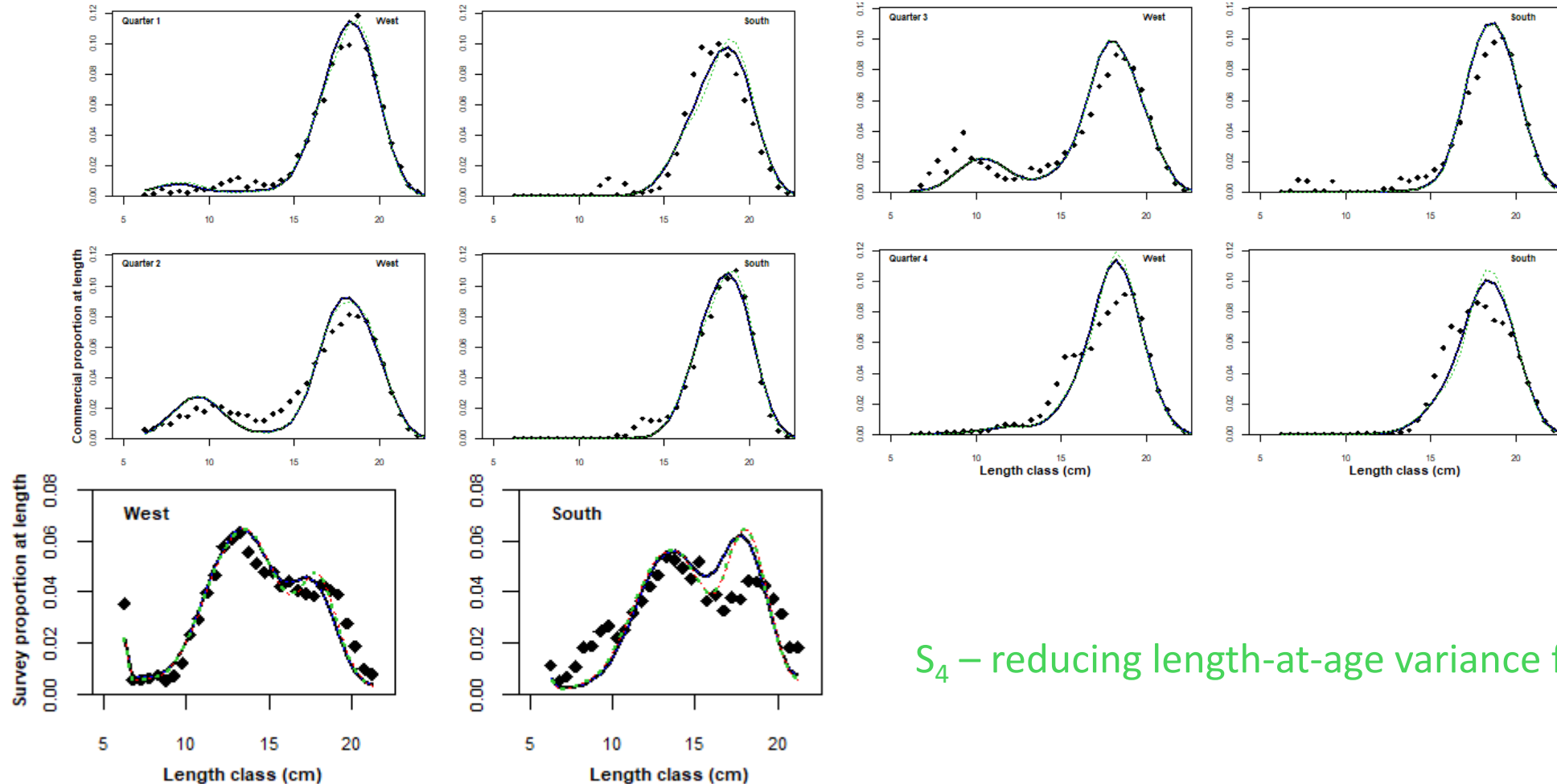


S_1 – downweight LF data all years

S_2 – downweight LF data 2018-2023

S_3 – downweight survey LF data 2018-2023

Improving fit to survey LF – less variance in length-at-age distribution?



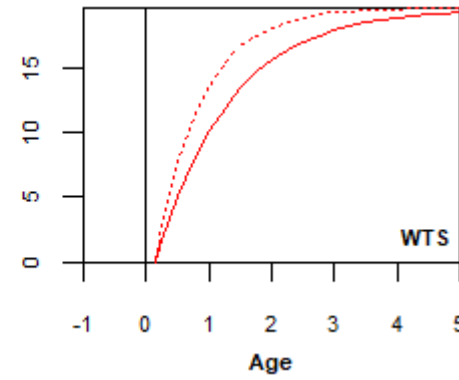
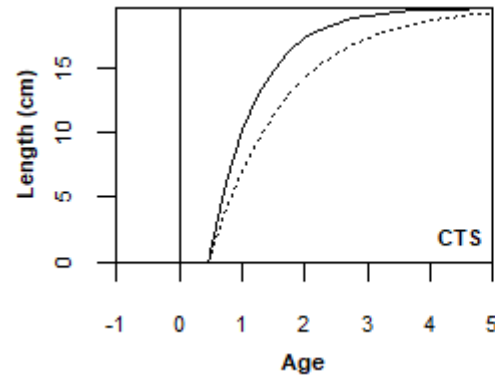
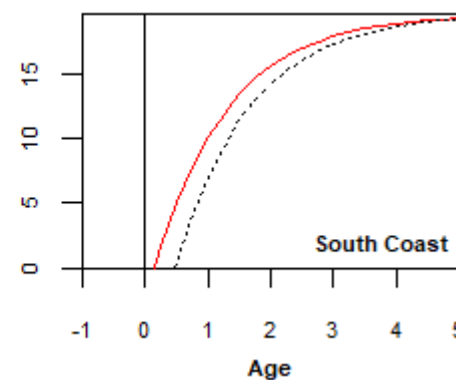
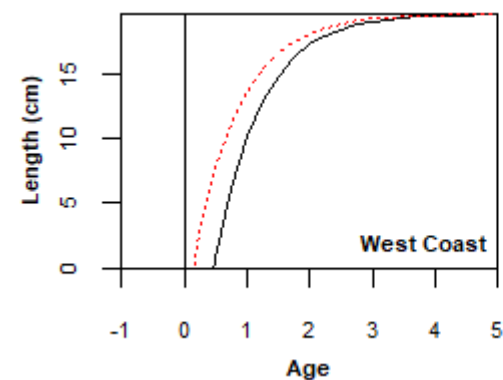
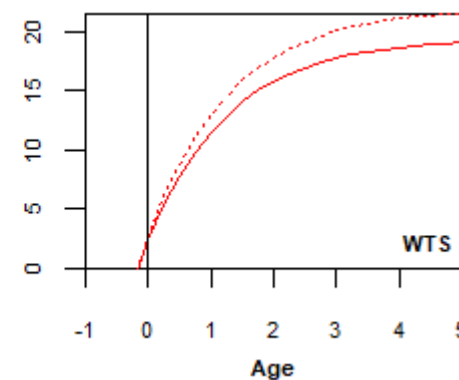
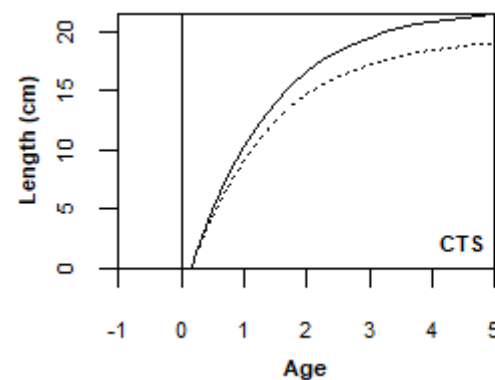
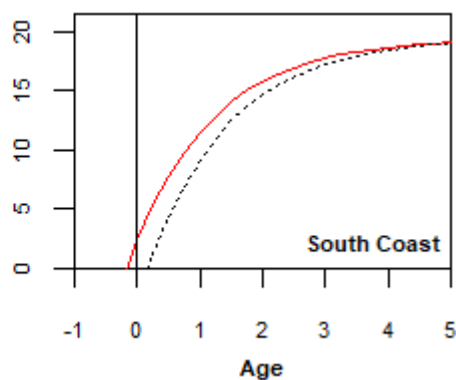
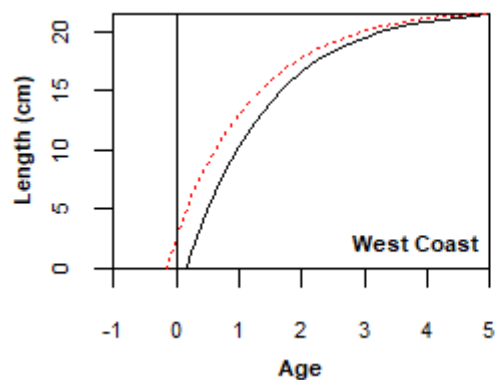
S_4 – reducing length-at-age variance for ages 2+

FISHERIES/2024/NOV/SWG-PEL/41 tested restrictions on variance of other ages too

FISHERIES/2024/NOV/SWG-PEL/41 tested decreasing selectivity at larger lengths

Estimating the growth curve

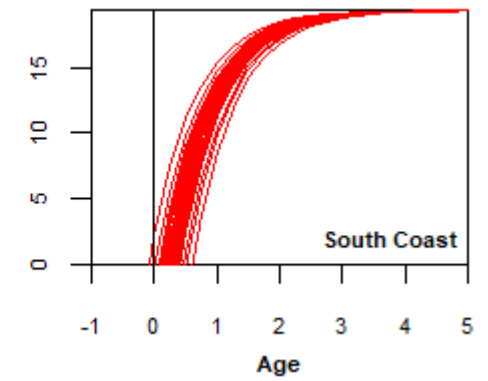
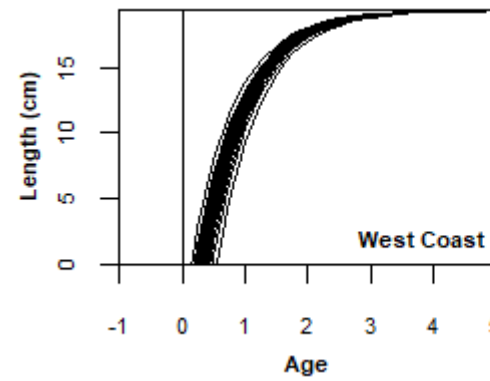
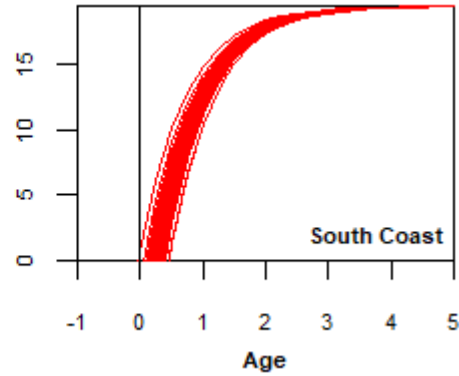
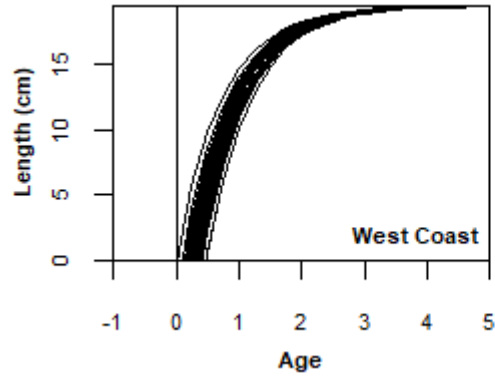
Estimating L_∞ (with t_0 and L_1 and constraint of $\kappa_w \leq \kappa_s$) instead of κ (with t_0 and L_1 and constraint of $L_{w,\infty} \leq L_{s,\infty}$) – same result



S_5 – no $L_{w,\infty} \leq L_{s,\infty}$ constraint

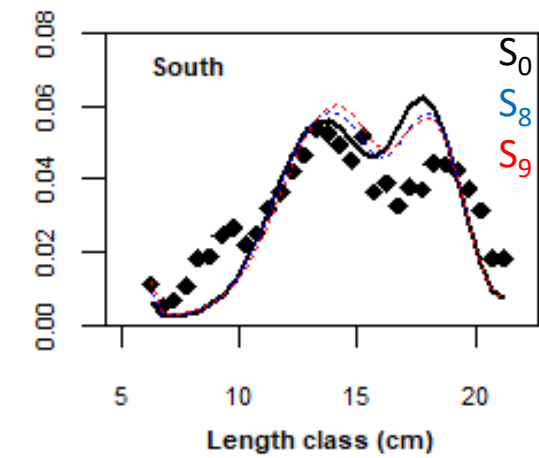
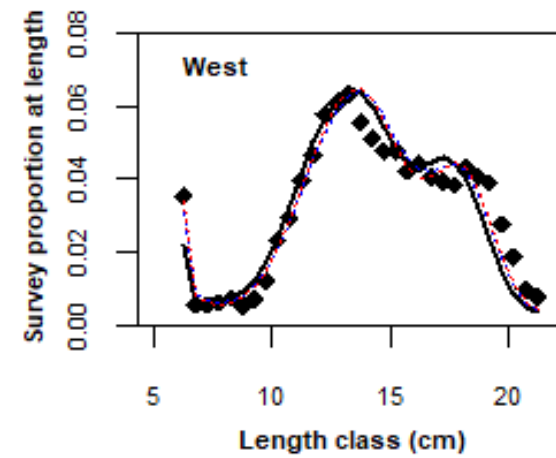
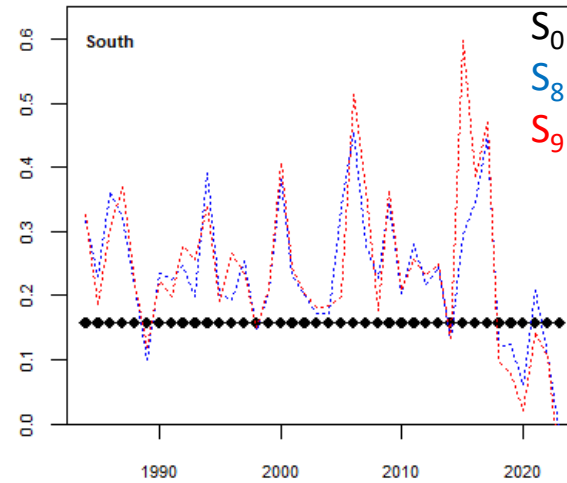
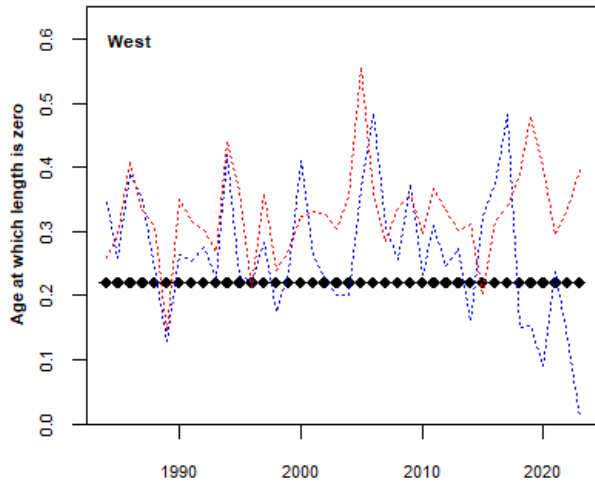
S_7 – no $\kappa_w \leq \kappa_s$ constraint

Earlier/later peak recruitment



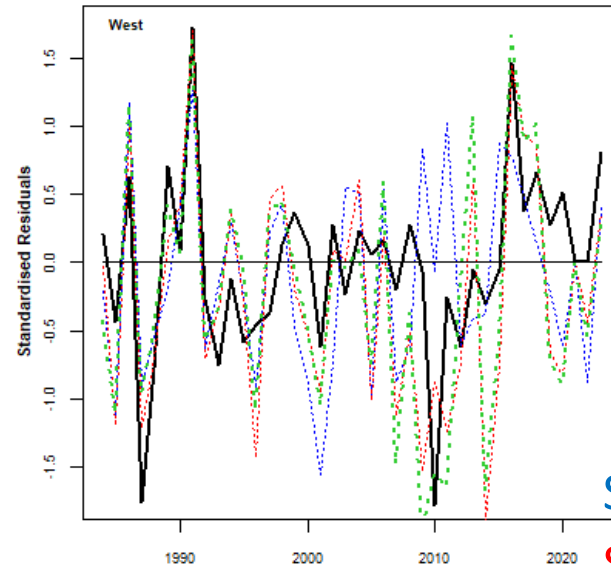
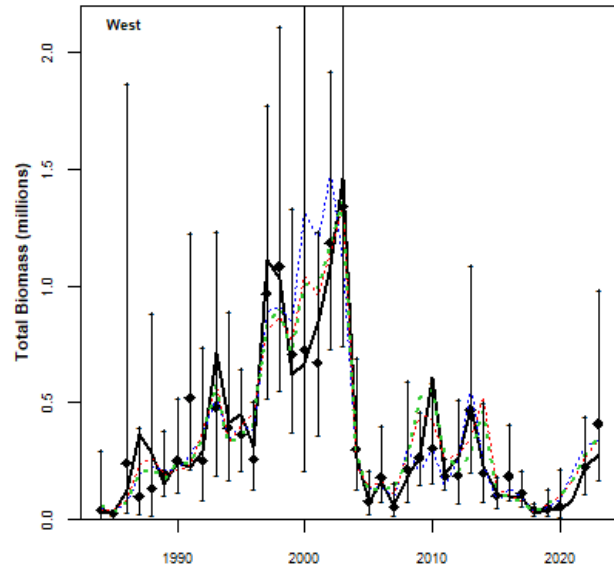
$$S_8 - \eta_{c,y}^t = \eta_y^t \sim N(0, 0.2^2)$$

$$S_9 - \eta_{c,y}^t \sim N(0, 0.2^2)$$

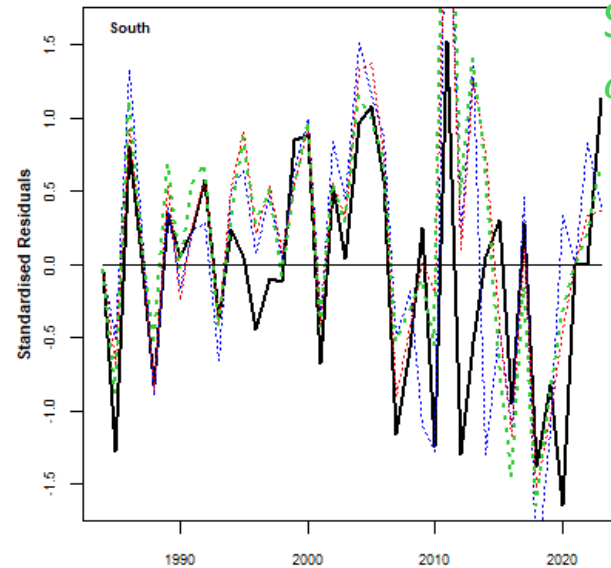
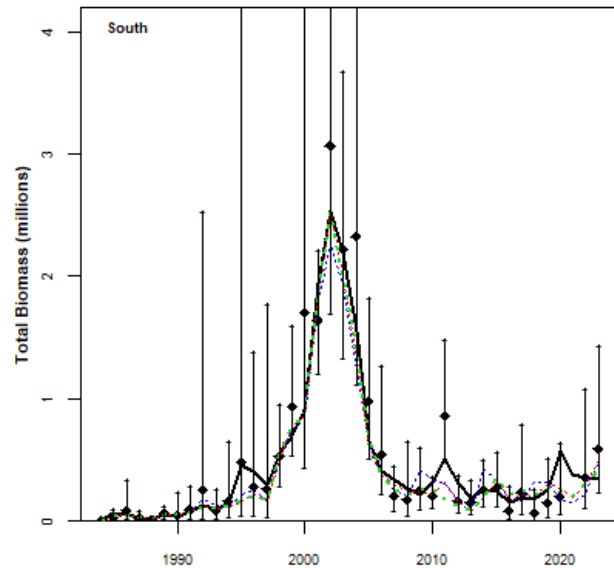


Substantial improvement in fit, particularly to survey LFs

Including parasite prevalence-at-length



Some loss in fit to other data
Nov 2020 biomass on SC estimated to be closer
to observation

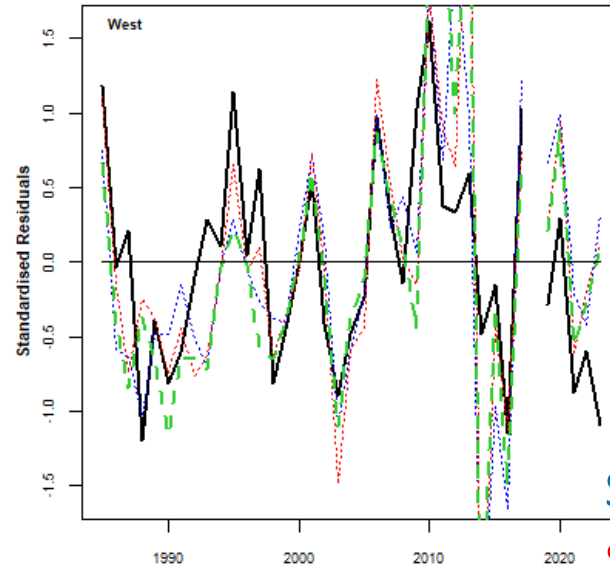
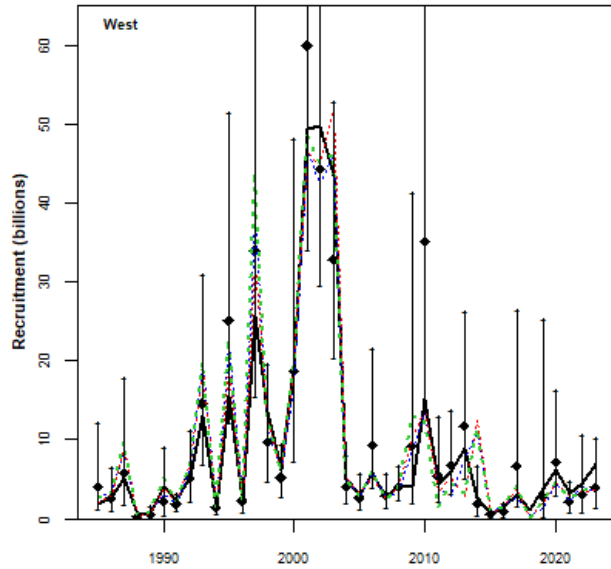


S_{10} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.5, 0.5)$

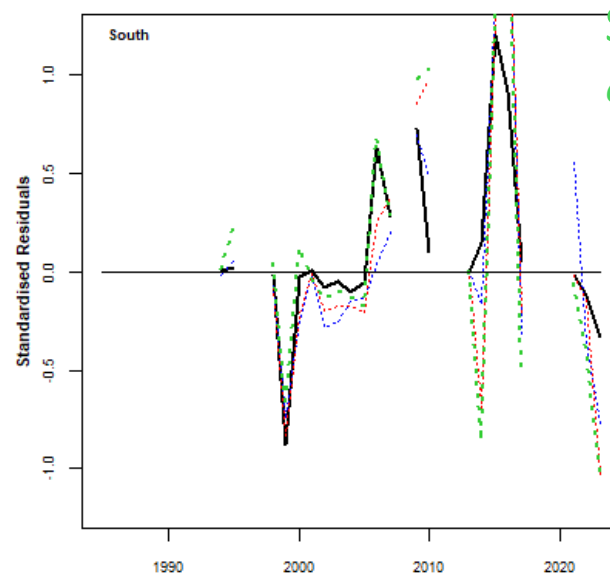
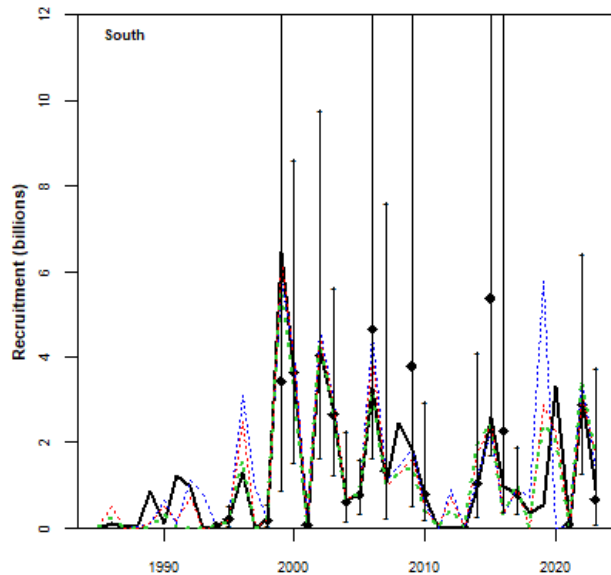
S_{11} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$

S_{12} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$,
 $\delta \sim U(0.005, 1)$ instead of $\delta \sim U(0.005, 5)$ and restricting $S_{50} \leq 8$

Including parasite prevalence-at-length



Some loss in fit to other data

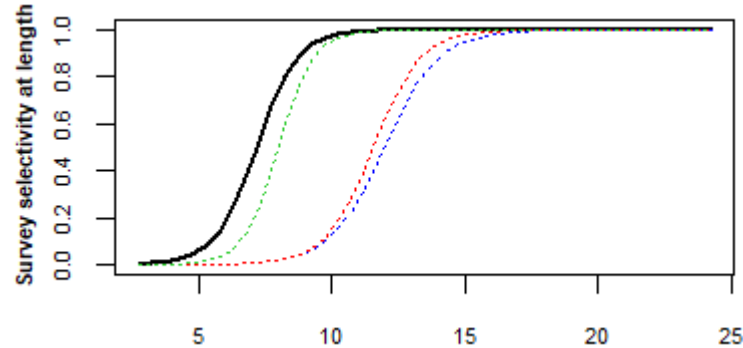


S_{10} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.5, 0.5)$

S_{11} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$

S_{12} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$, $\delta \sim U(0.005, 1)$ instead of $\delta \sim U(0.005, 5)$ and restricting $S_{50} \leq 8$

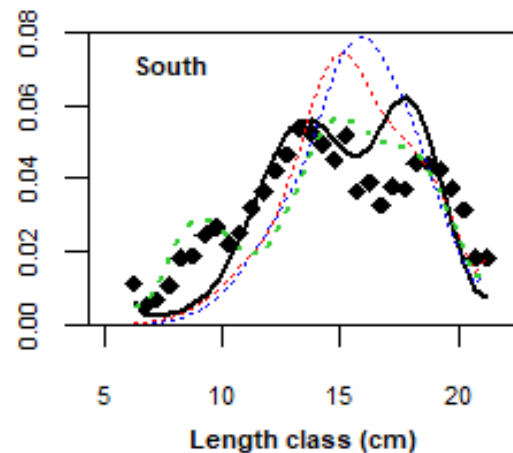
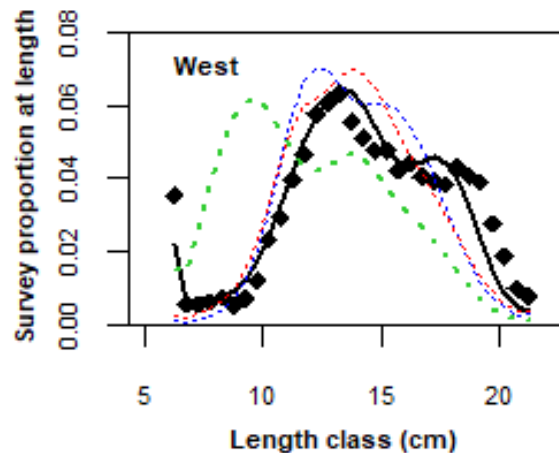
Including parasite prevalence-at-length



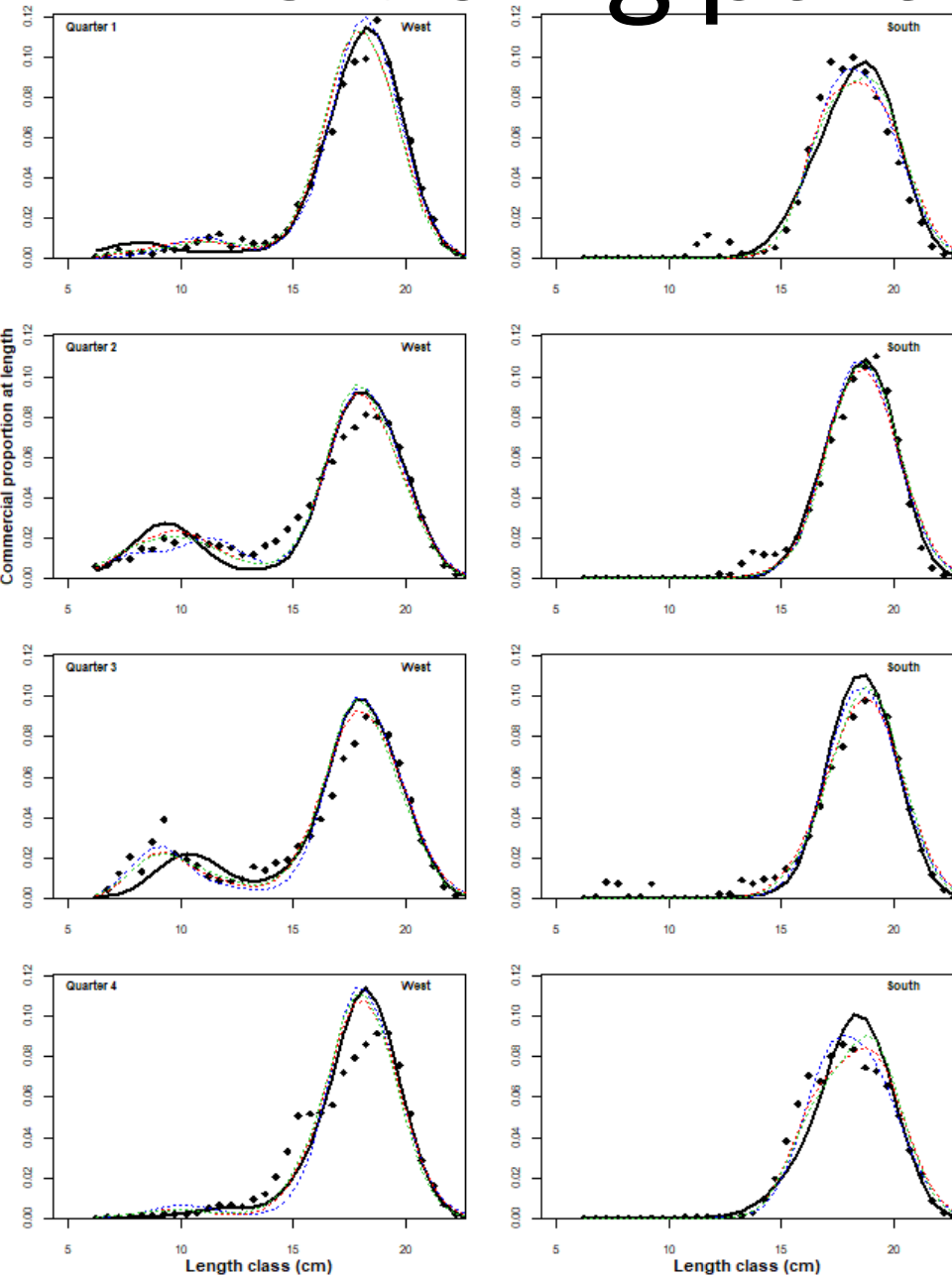
S_{10} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.5, 0.5)$

S_{11} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$

S_{12} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$, $\delta \sim U(0.005, 1)$ instead of $\delta \sim U(0.005, 5)$ and restricting $S_{50} \leq 8$



Including parasite prevalence-at-length

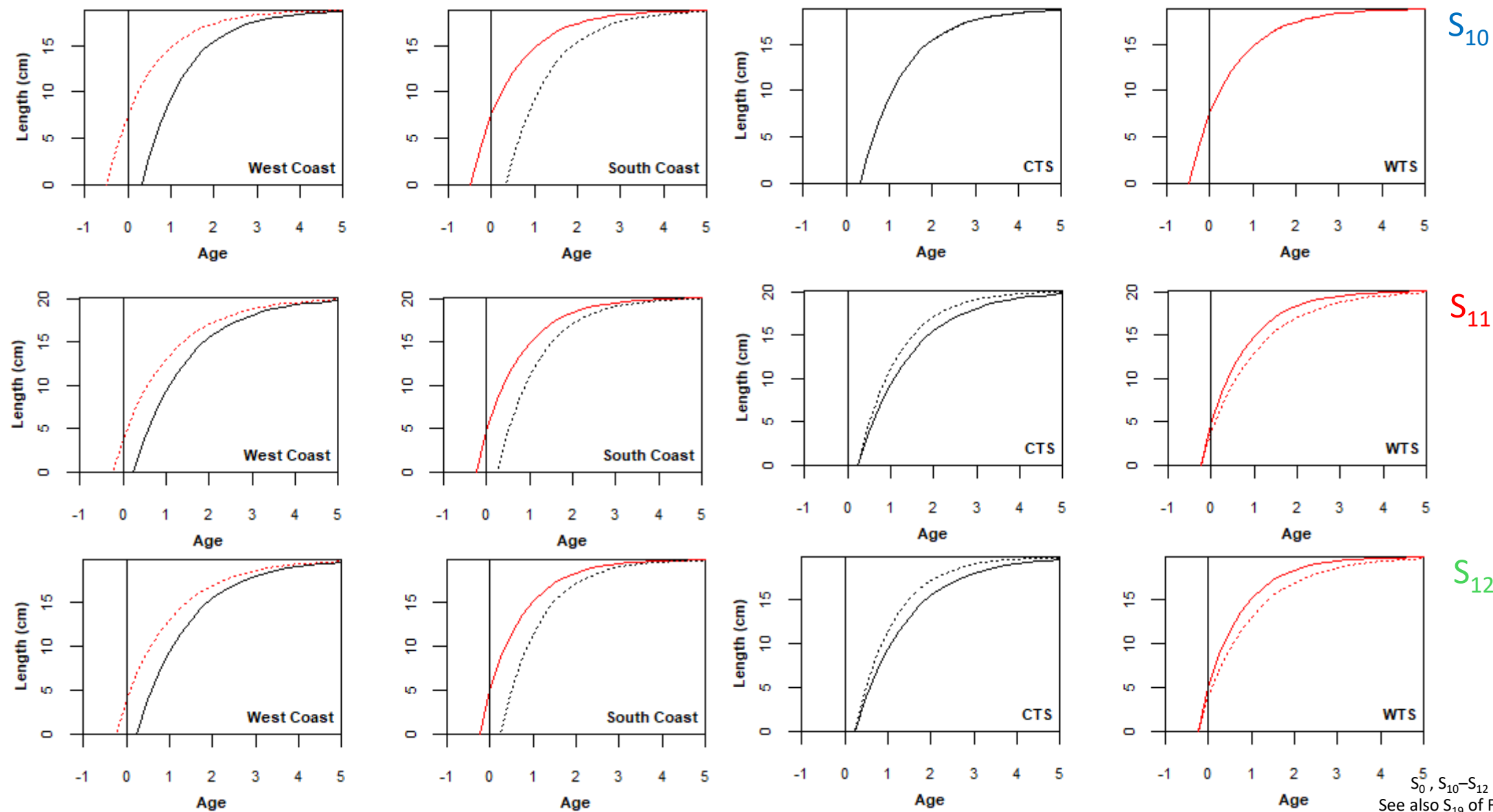


S_{10} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.5, 0.5)$

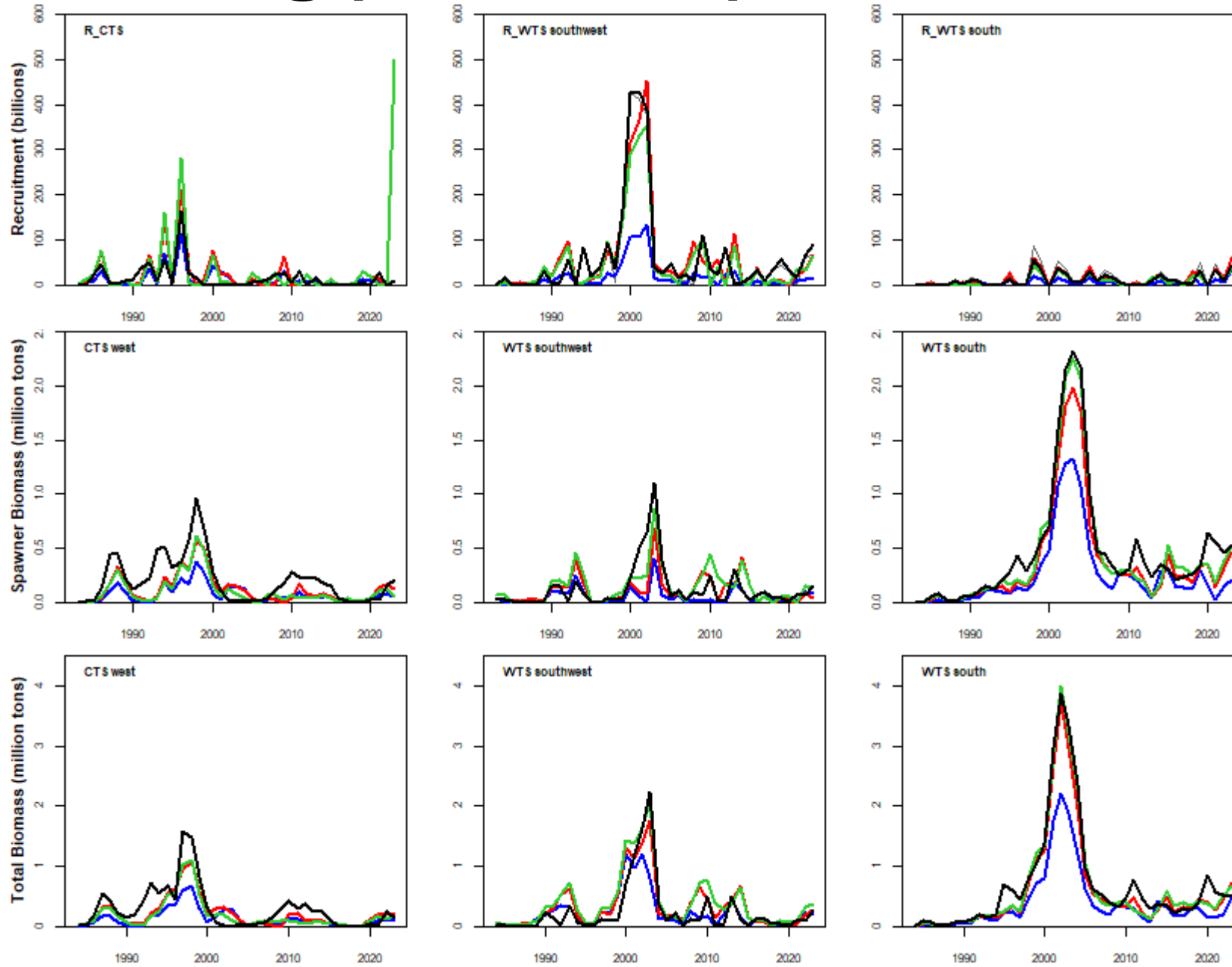
S_{11} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$

S_{12} – including parasite data with $t_{0,s}^{WTS}, t_{0,w}^{CTS} \sim U(-0.25, 0.25)$,
 $\delta \sim U(0.005, 1)$ instead of $\delta \sim U(0.005, 5)$ and restricting $S_{50} \leq 8$

Including parasite prevalence-at-length

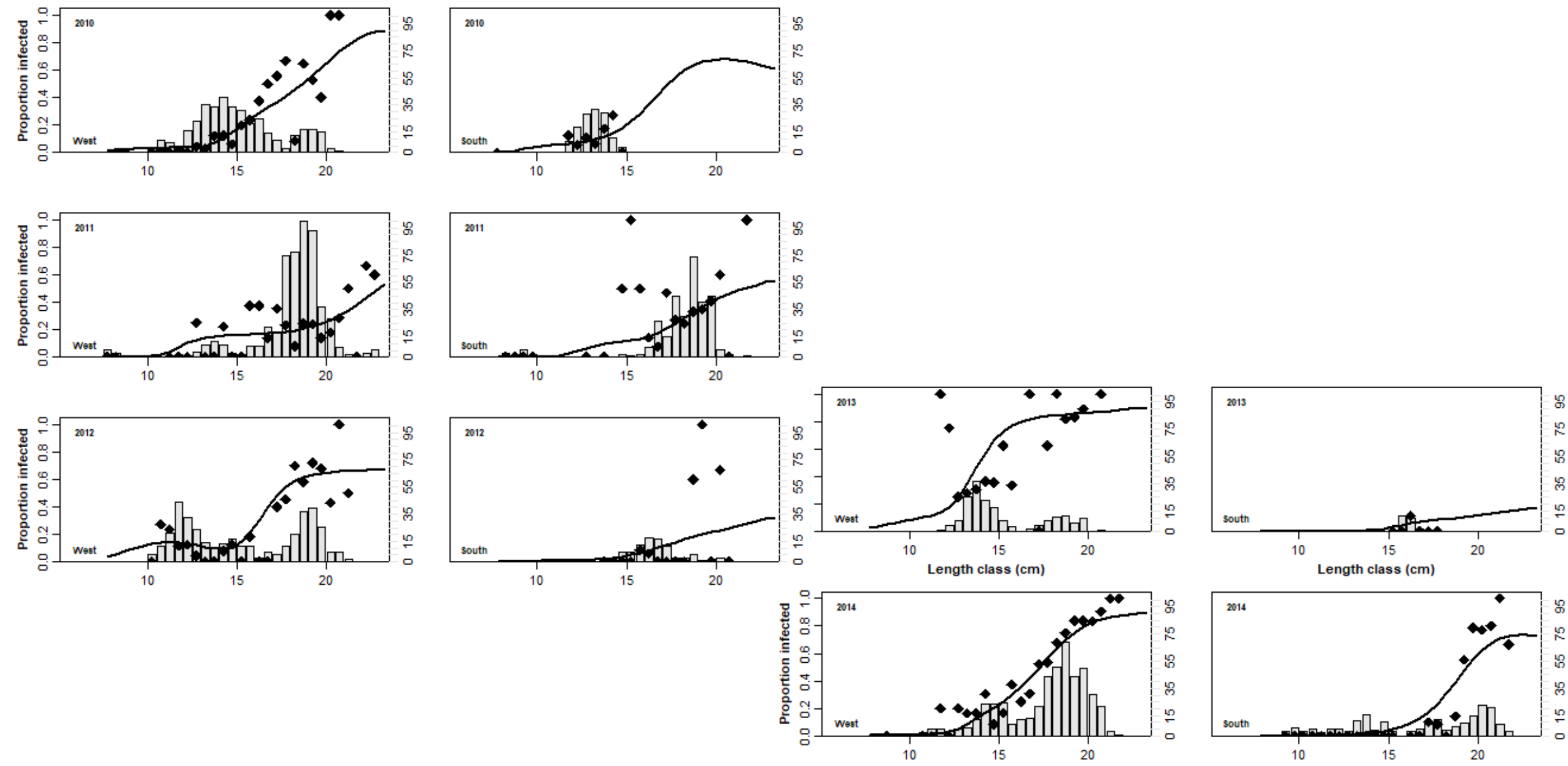


Including parasite prevalence-at-length

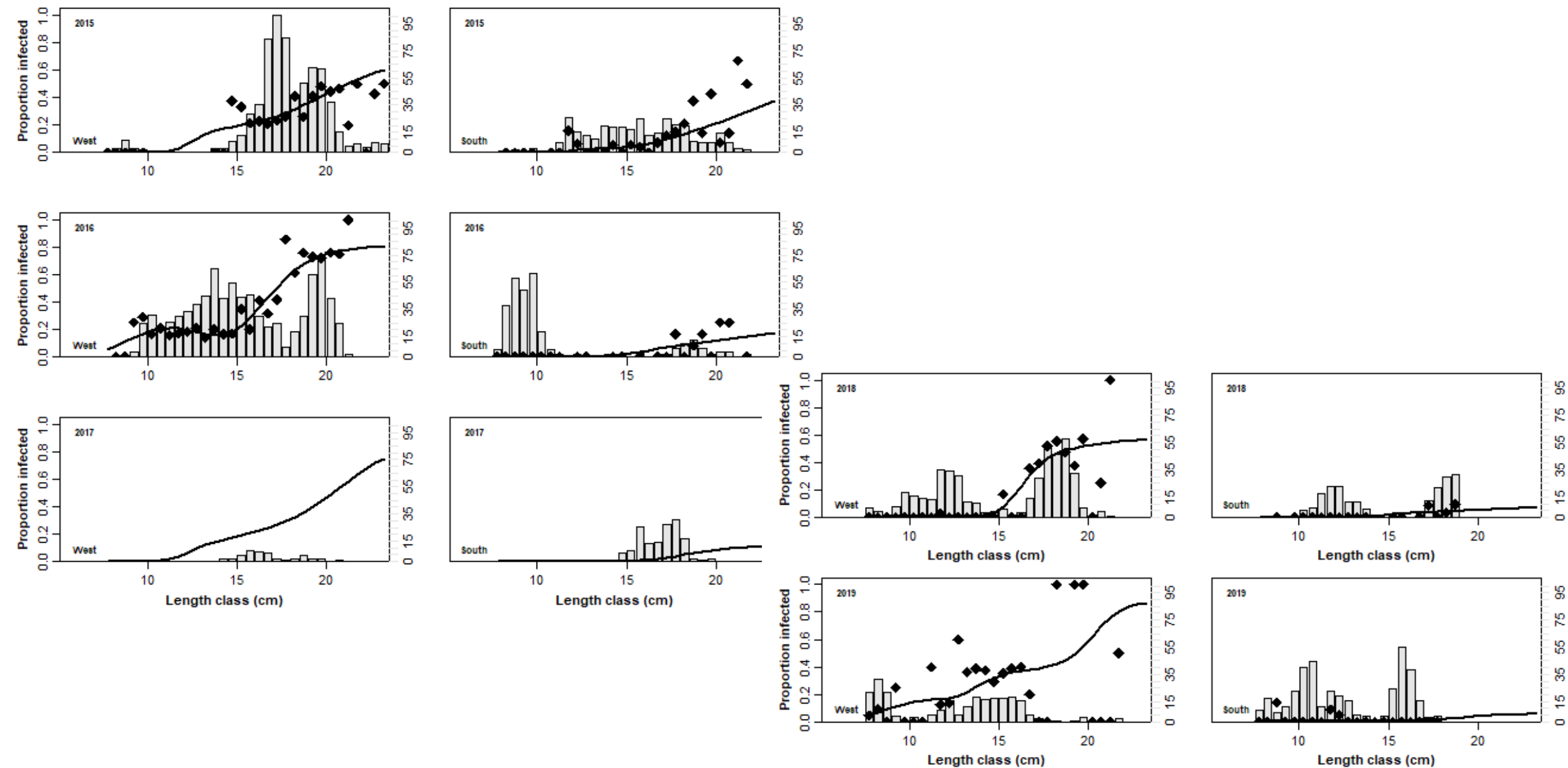


S_0
 S_{10}
 S_{11}
 S_{12}

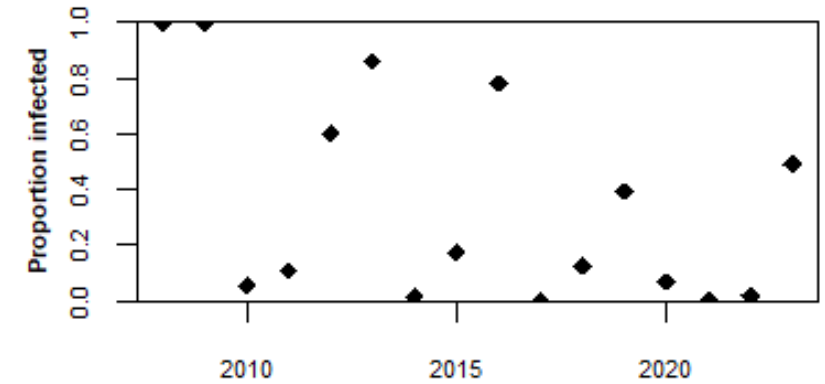
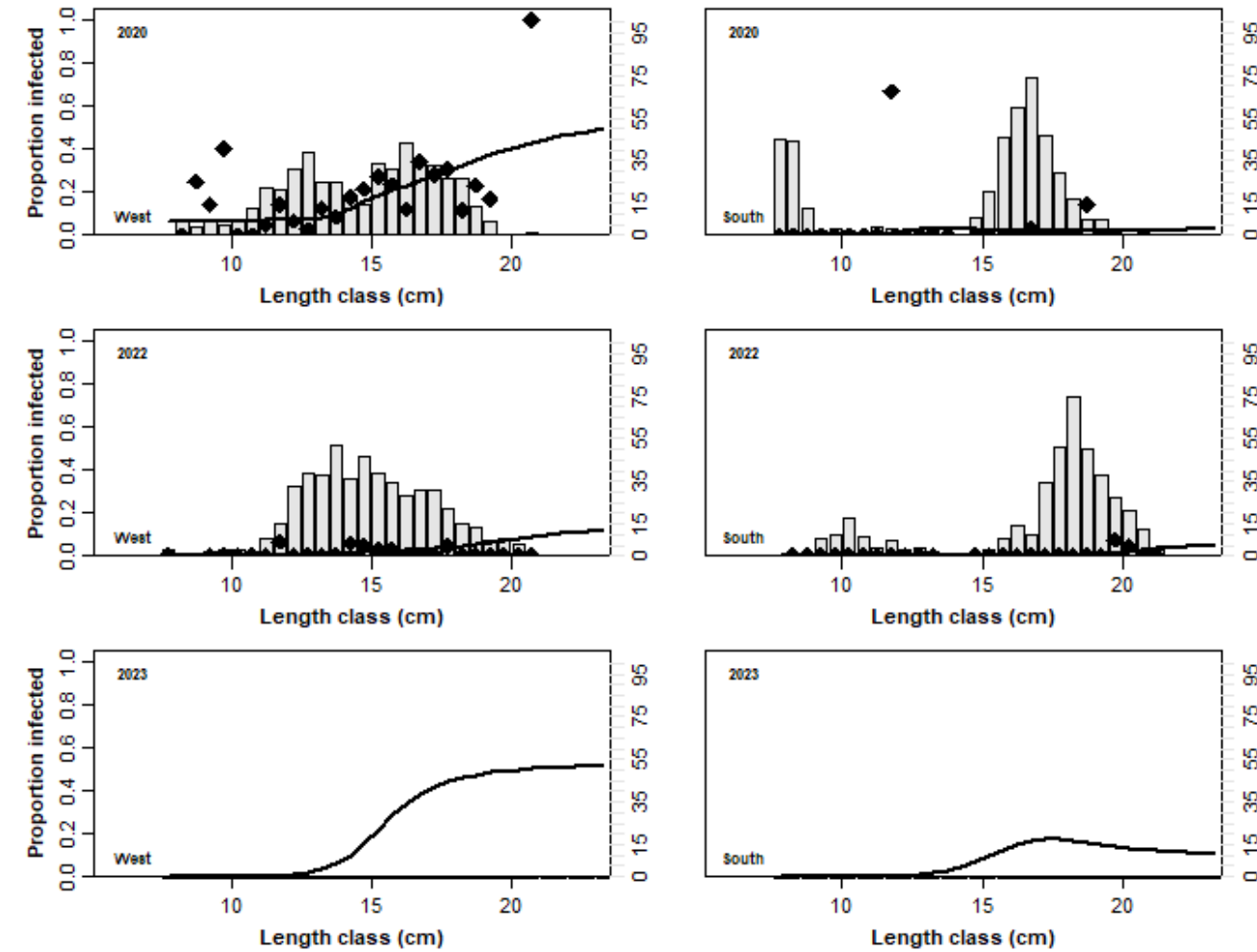
Including parasite prevalence-at-length



Including parasite prevalence-at-length

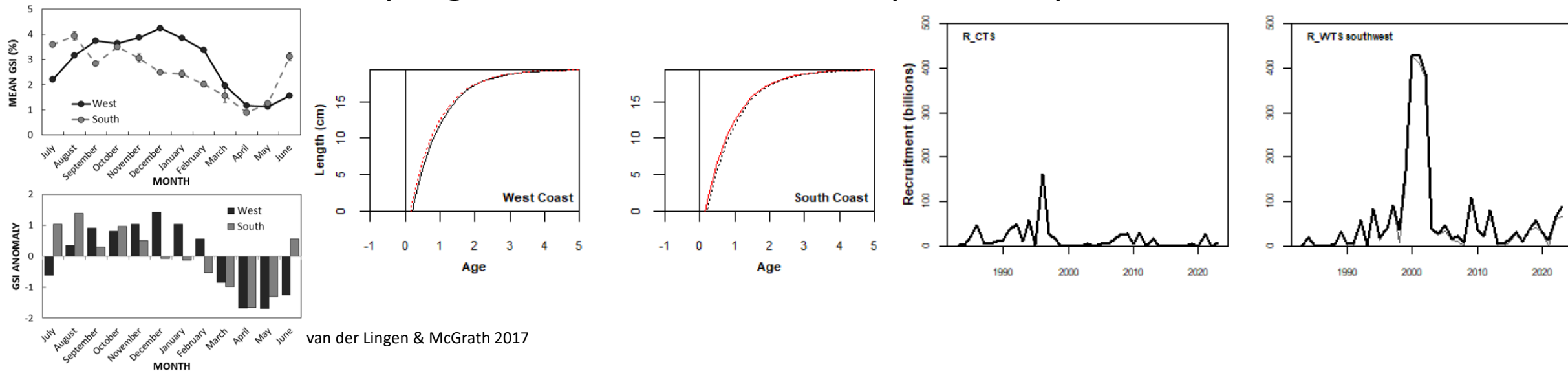


Including parasite prevalence-at-length



Concerns?

- Peak recruitment on SC ~ late November is much later than the June – November period indicated by the GSI of sardine sampled from commercial catches off the south coast; GSI indices calculated using data over ~20 years
- WTS recruitment to the WC is estimated to be substantial in many years, often larger than CTS. If WTS recruitment is substantial, spawning of WTS off the WC should be sufficiently significant to have been picked up in GSI indices

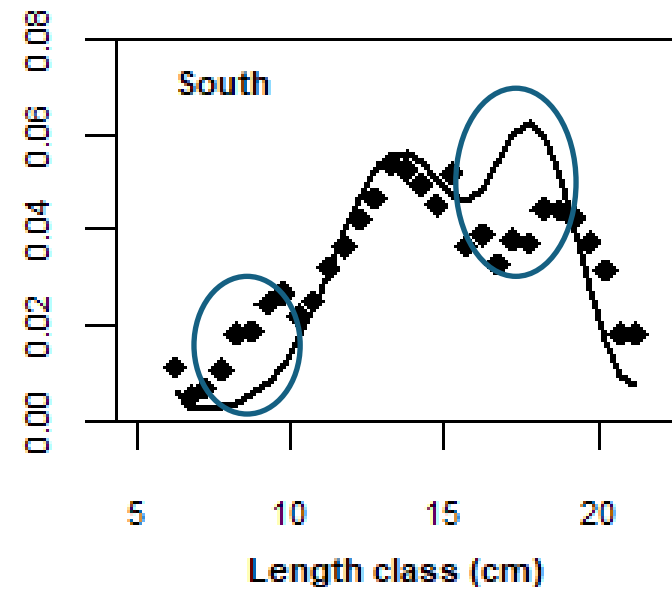
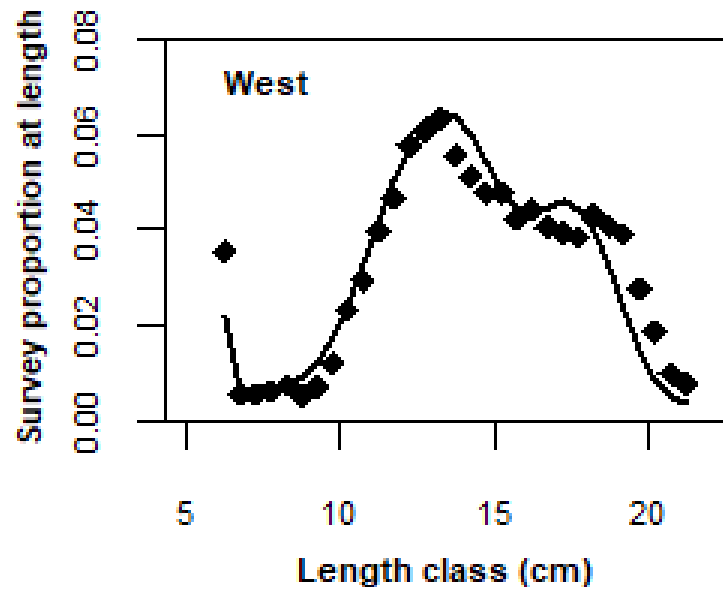


van der Lingen & McGrath 2017

Figure 1: Mean ($\pm$ standard error) monthly GSI (upper plot) and GSI anomaly (= mean monthly GSI – overall average GSI) for sardine caught off the West (dark circles/histograms and solid line) and South (light circles/histograms and dashed line) coast over the period 1995-2015.

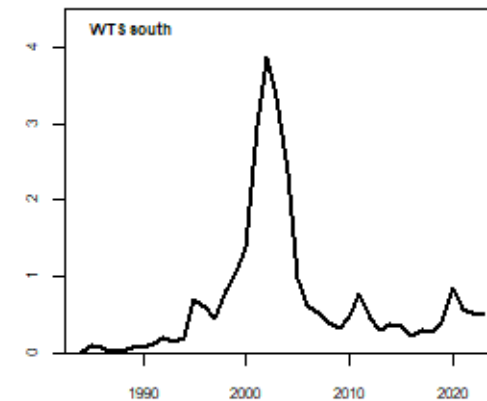
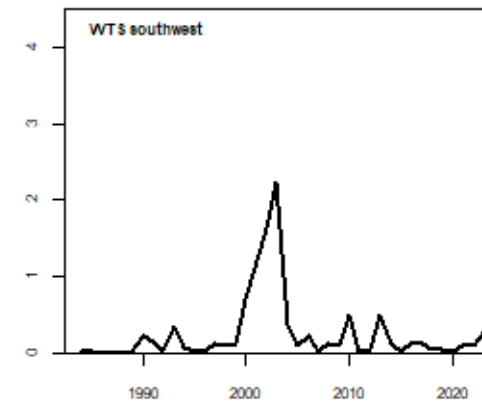
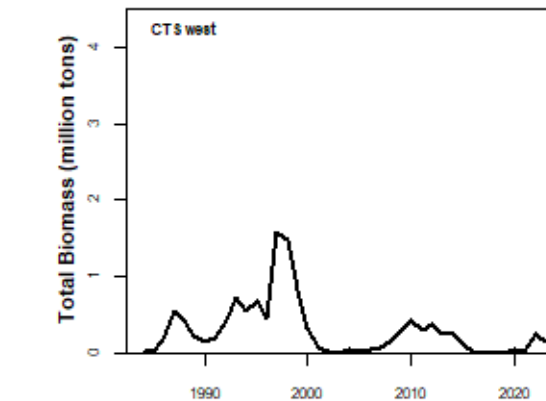
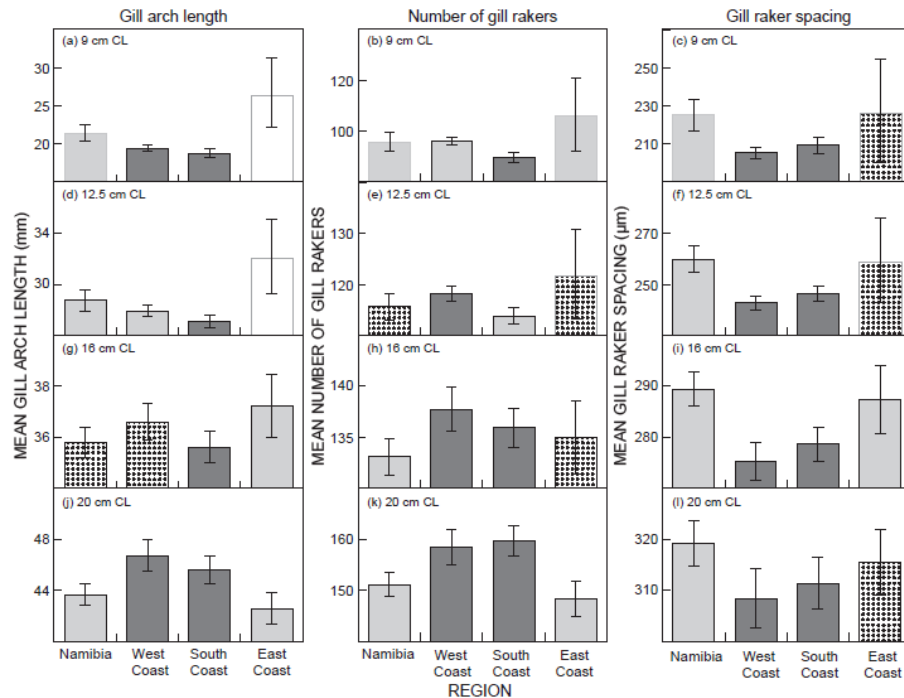
Concerns?

- More large fish are predicted off the SC than that observed in November surveys



Concerns?

- WTS (spawner) biomass off the WC is estimated to have been a substantial component of WC biomass not only during pulse years, but a number of other years. Does this contradict morphometric (probably not) and meristic (maybe) differences observed between coasts.

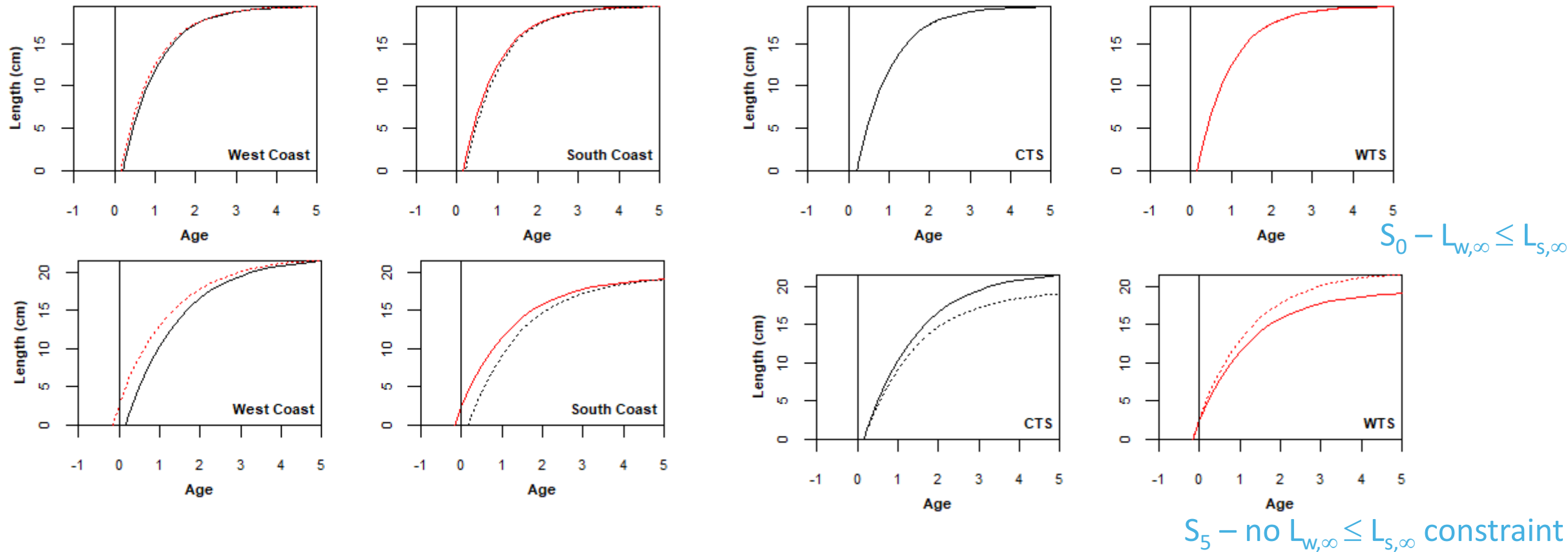


Idris et al. 2016

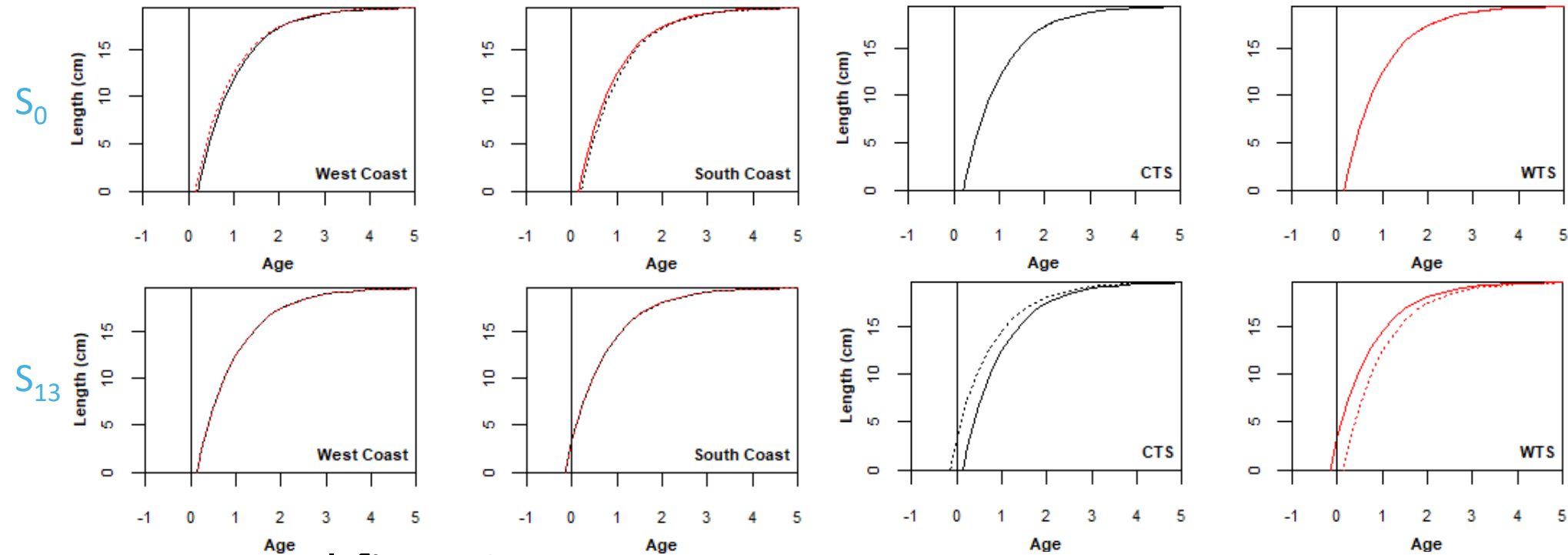
Figure 6: Results of the multivariate GLM showing predicted mean gill arch lengths, numbers of gill rakers and gill raker spacings for sardine of different sizes (approximate ages) in the four regions; (a, b, c) 9 cm CL (6 months old), (d, e, f) 12.5 cm CL (1 year old), (g, h, i) 16 cm CL (2 years old) and (j, k, l) 20 cm CL. Error bars denote $\pm 95\%$ CI. Different solid shadings (none, light or dark) indicate statistically significant differences between groups, patterned bars indicate the region could be assigned to more than one group and histograms with a pale outline for smaller (9 and 12 cm CL) fish represent model extrapolations beyond the range of the data

Concerns?

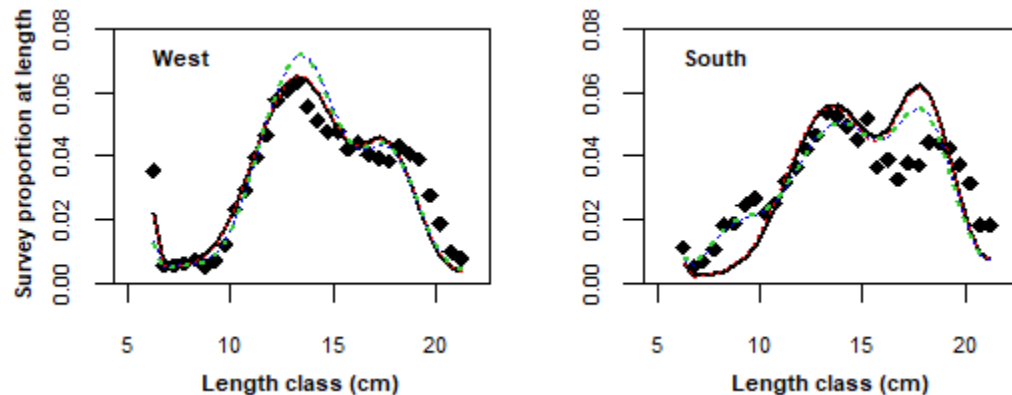
- Additional constraints are required to obtain growth curves that align with observations of faster early growth off the SC



Timing of WTS recruitment off the WC



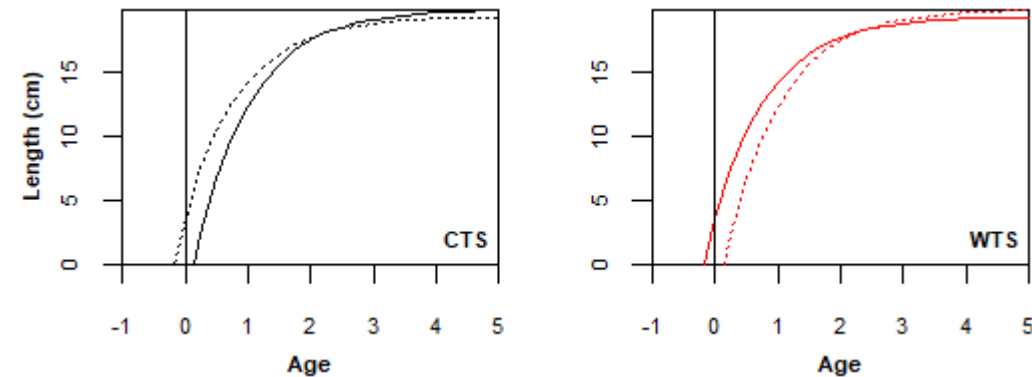
Improved fit to SC survey LFs



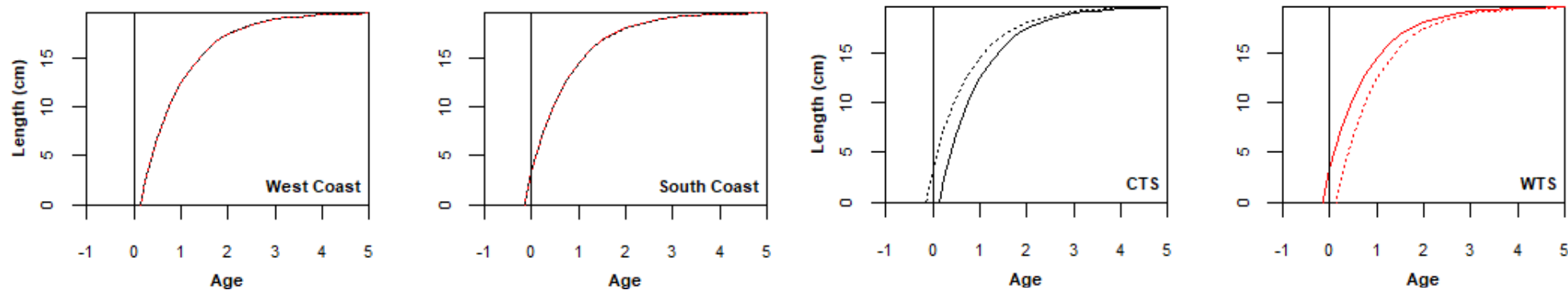
$$S_{13} - t_{0,w}^{WTS} = t_{0,w}^{CTS}$$

$$S_{11} - R_{c,y}^{WTS} \leq 50$$

$$S_{12} - R_{c,y}^{WTS} \leq 50$$



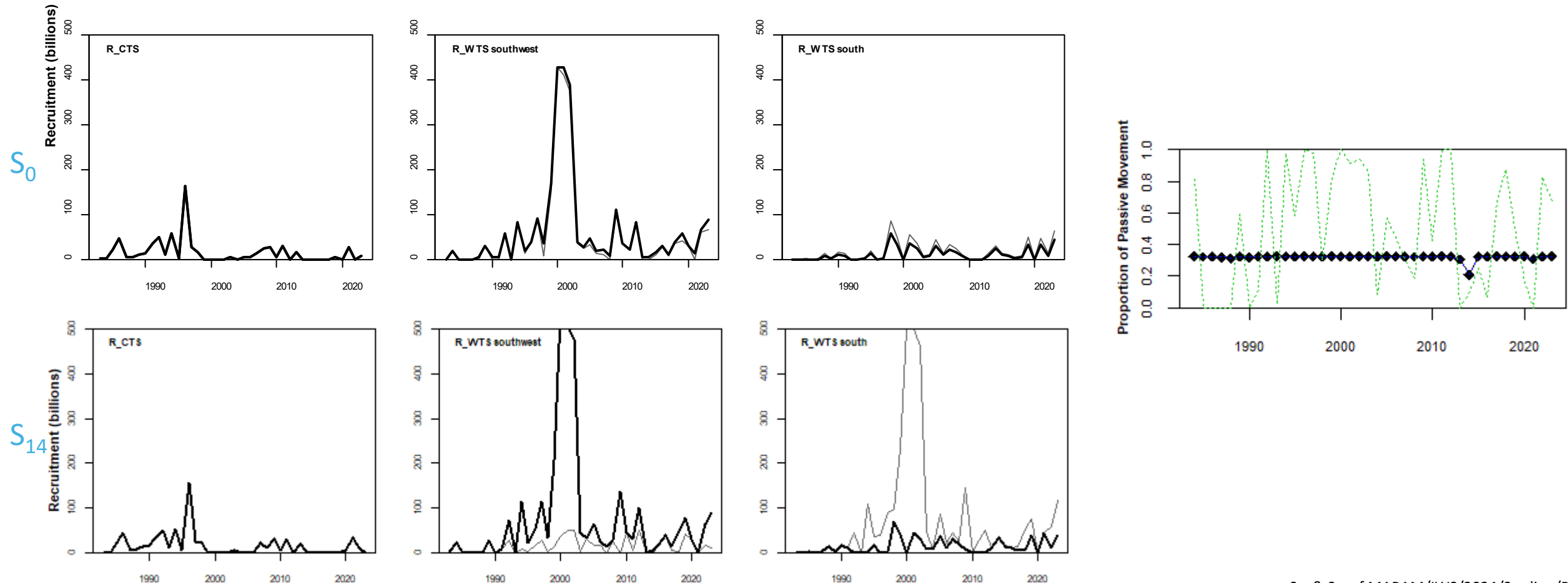
Timing of WTS recruitment off the WC



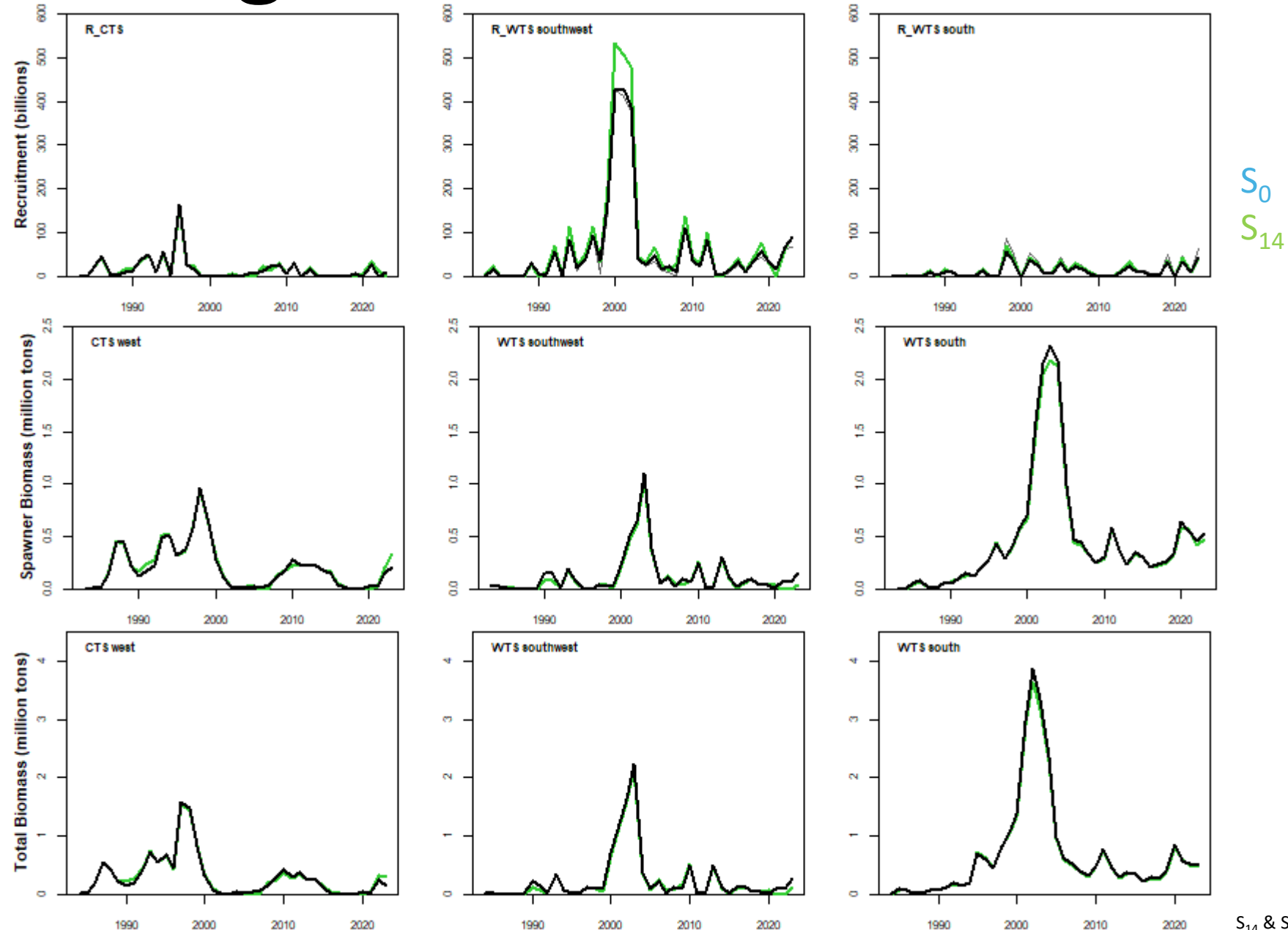
- NB current model formulation assumes WTS spawned off the south coast, but passively moved to the west coast as recruits have same peak recruitment time as recruits resulting from west coast spawning

Restricting WTS recruitment off the WC

- $R_{c,y}^{WTS} \leq 50$ (Arbitrary choice for this example)
- If the prior on passive movement is $U(0,1)$, the fit to the data is *slightly* better

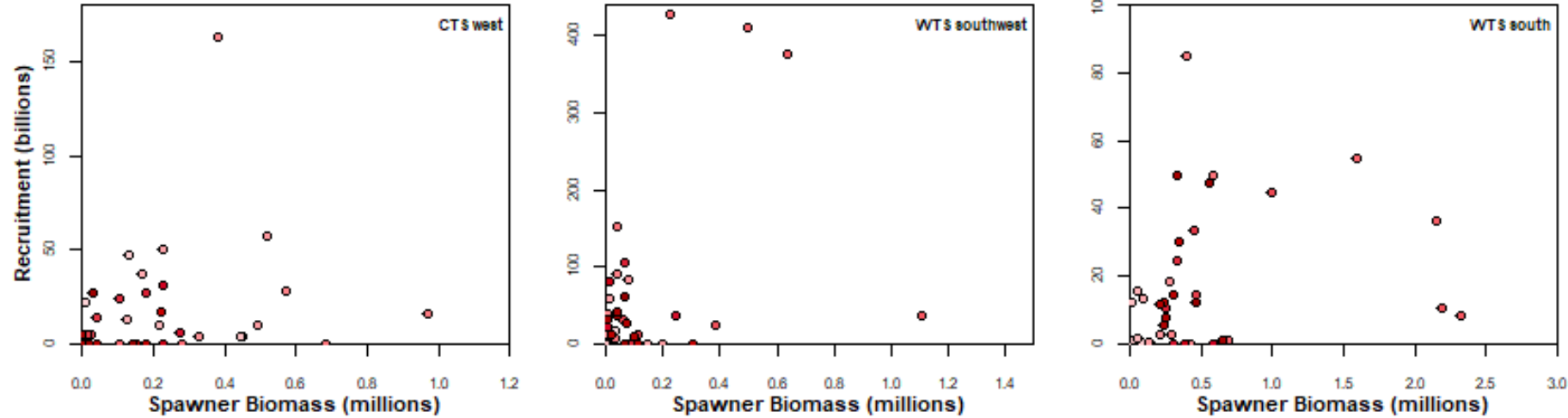


Restricting WTS recruitment off the WC

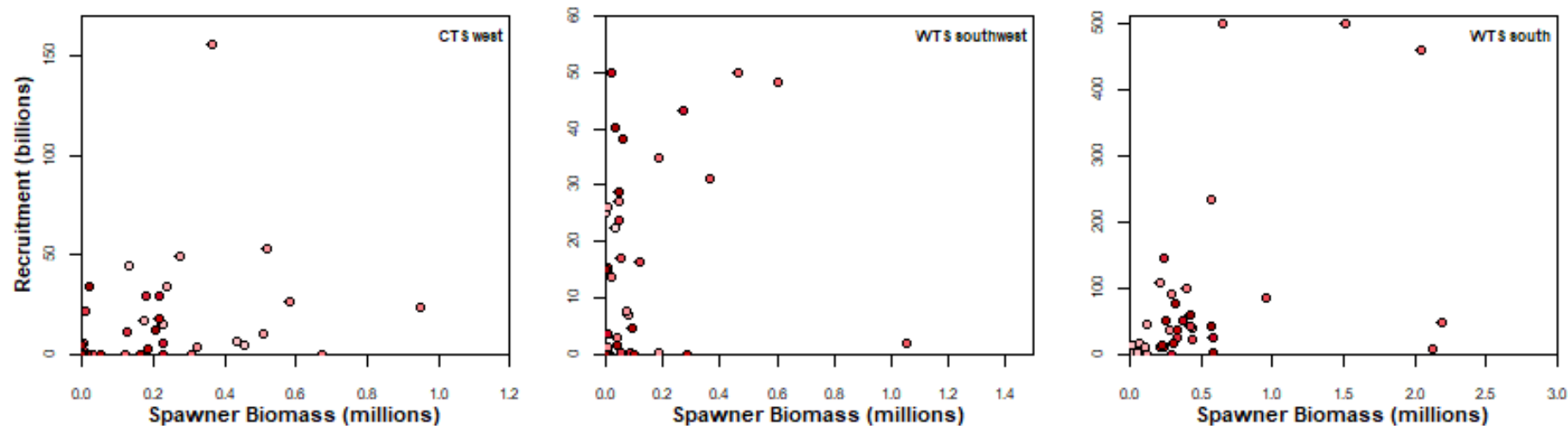


Restricting WTS recruitment off the WC

S_0



S_{14}



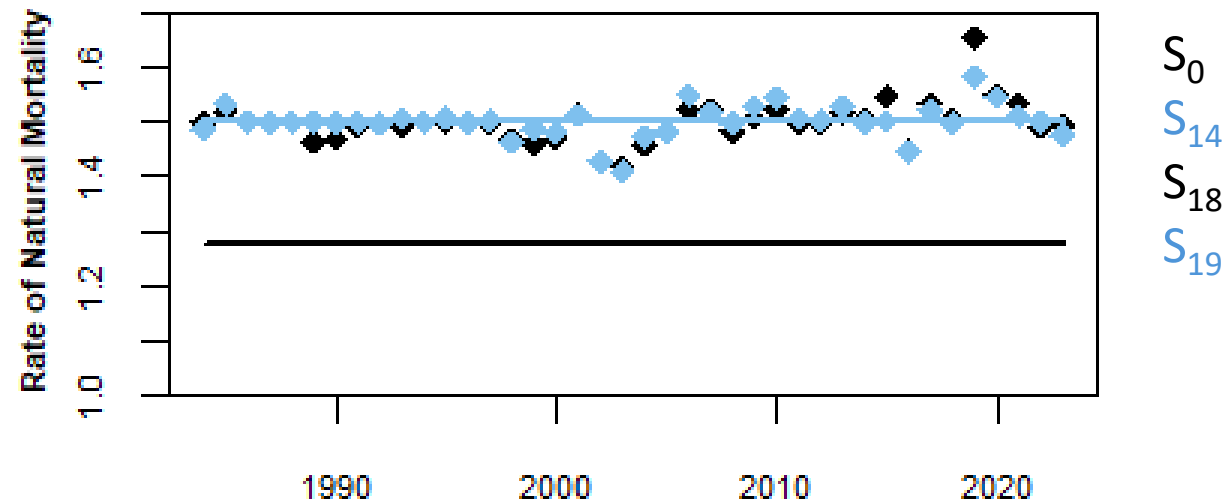
Should we e.g.
consider an
informative prior on
WTS recruitment off
the WC?

Single WTS recruitment

- If a single timeseries of WTS recruitment is estimated which is subsequently split between the west and south coasts (this annual proportion then incorporating both coast-specific recruitment and passive movement), the net recruitment is similar to S_0 / S_{13} and the fit to the data are thus similar to S_0 / S_{13}

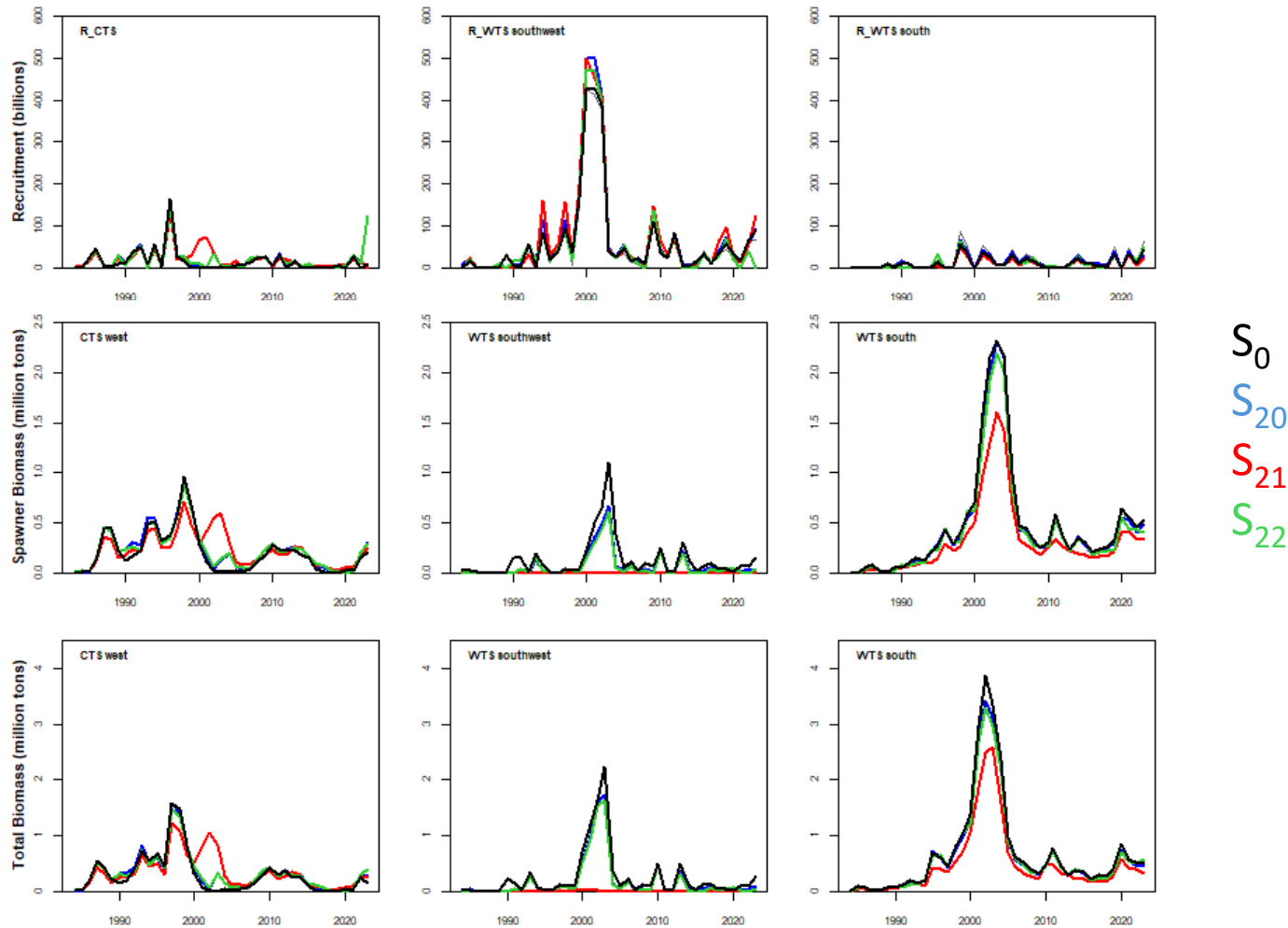
Variable WTS juvenile M off West Coast

- A relatively small improvement in fit to data
- Possible that any 'favourable environment' would impact both juvenile and adult WTS off the West Coast?



Less older WTS off West Coast

- S_{20} , S_{22} – all WTS move by age 2 (v S_0 , S_{15}) - fit slightly worse
- S_{21} – all WTS move by age 1 (v S_0) - fit worse



Less older WTS off West Coast

